Modern Force and Coercive Backfire

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Abstract. Modern force enables states to conduct subwar operations—bounded uses of force short of general war. Does possessing the capability to conduct such operations improve coercion? We study this question in a parsimonious bargaining model where the coercer chooses its demand. Coercion requires both threat and assurance: resistance must be costly, and compliance must prevent further harm. By replacing war, the operation can reduce harm; by replacing settlement, it can create harm. The capability helps when it widens the gap between punishment after resistance and harm not prevented by compliance, and hurts when it narrows it. The target can therefore accept less, and the coercer's expected payoff can fall even where the operation reduces war. Where the operation cannot compensate for the smaller concession, every coercer type is weakly worse off, and types that would otherwise have settled instead choose war.

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Introduction

Large interstate wars have become less frequent, and whether that pattern reflects a real decline in the propensity for war remains contested (Braumoeller 2019; Pinker 2011). Both sides of the debate count wars. Modern force increasingly enables states to conduct subwar operations—bounded uses of force intended to remain short of general war. Armed drones, for example, combine remote operation, persistent surveillance, and precision strike while reducing personnel exposure (Horowitz 2020; Zegart 2018). Violence nevertheless remains common in international politics (Davies, Pettersson, and Öberg 2022). States seize positions incrementally, intercept shipping, launch stand-off strikes, conduct cyber operations, and arm proxies while trying to remain below the threshold of general war (Gannon et al. 2024; Kenkel and Schram 2025; Schram 2021). This juxtaposition motivates the paper's question: how does acquiring the capability to conduct such operations affect interstate bargaining? Does it broaden or narrow the prospects for peace?

The capability can either broaden or narrow the prospects for peace. Coercion requires both a credible threat against resistance and a credible assurance that compliance prevents further harm (Cebul, Dafoe, and Monteiro 2021; Kydd and McManus 2017; Pape 1996; Schelling 1966). A subwar operation can support peace by replacing war after resistance, or strengthen the threat by replacing settlement there. Yet if the operation remains profitable after compliance, it can replace settlement and weaken assurance. The target then accepts a smaller demand whenever this loss of protection outweighs any increase in punishment after resistance. We call the resulting bargaining performance *coercive ability*: the expected portion of an accepted demand that remains in the coercer's hands without general war. The demand x is the price the target agrees to pay; coercive ability discounts that amount by the probability that general war later reopens the dispute. It is neither the frequency of compliance nor the coercer's total payoff. Except in one limiting case, the target complies in both games, and what changes is how much it pays and whether the payment survives.

We establish this result in a parsimonious bargaining framework of interstate conflict built from two minimal extensions to a canonical crisis-bargaining model. The canonical model has an informed coercer choose a demand and the target comply or resist (Banks 1990; Fearon 1994; Schultz 1999). First, we let the coercer choose between settlement and general war after either response, rather than only after resistance. Second, we add a subwar operation that costs less than war to the same continuation menu. The benchmark incorporates the first extension; the operation game adds the second, enabling a clean formal experiment of the new capability's effect (Paine and Tyson 2020). We select the largest accepted demand in each game. Because the games differ only in the continuation menu, the comparison isolates how the capability changes the target's willingness to comply.

The model yields three main results. First, the capability can reduce coercive ability. In the benchmark, greater power raises punishment after resistance and supports a larger concession. With the capability, reach and relative cost determine harm after both responses. For weak coercers, the operation can add punishment where war is not credible and improve coercive ability. For stronger coercers, however, it can replace a more damaging war after resistance while remaining profitable after compliance. Coercive ability can therefore fall even while punishment remains higher; as power rises further, threat and assurance can weaken together. The target responds to punishment relative to post-compliance harm, so the accepted demand and war need not move with coercive ability. Modern technologies that lower the military cost of using force can therefore raise the political cost of settling disputes.

Second, the capability can increase war through the target's smaller concession. At a fixed demand, the operation can replace war for low-cost types and settlement for higher-cost types. When active after resistance, it never raises war at that history. Equilibrium demands, however, are not fixed. If the target accepts less, types that would have honored the benchmark settlement receive too little and choose war after compliance. The direct menu effect and the demand response therefore move war in opposite directions: the operation replaces war for one set of types, while the reduced concession induces war among another. War rises when the second response moves the post-compliance cutoff farther. This increase can persist even when no type uses the operation after compliance. The operation weakens punishment after resistance, the target pays less, and the smaller payment sends types that never use the operation to war.

Third, the target's smaller concession can cost the coercer more than the operation can provide. At any given demand, access to the operation cannot hurt; each type can choose as before. But the target anticipates the additional option and offers less. Once the lost concession exceeds the operation's full gain, no coercer type gains. Those who fight in both games receive the same payoff, while every type that would have settled without the capability loses. Some now choose war, some conduct the operation, and some still settle for less. The operation remains optimal for its users; losses arise from the worse bargain, not a bad continuation choice. Even types that never conduct the operation can lose from its availability. The coercer would be better off if it could commit not to use the operation after compliance (Schelling 1960).

The paper first contributes to formal research on subwar operations. Existing models show how limited retaliation can make future punishment credible when a threat of massive retaliation is not (Powell 1989); how hassling a rising power can avert preventive war (Schram 2021); and how a challenger can withhold its most damaging instrument when a defender's escalation threat is credible (Gannon et al. 2024). Slantchev (2003a) establishes the capacity to impose costs short of victory as a dimension of power distinct from the capacity to win, and argues that this capacity need not track the capabilities ordinarily used to measure force (Slantchev 2011). A related literature treats war as a bargaining process in which fighting and negotiation can continue together (Slantchev 2003a; 2003b). These models illuminate mechanisms absent here, but not all share the same one. In complete-information process models, force changes continuation payoffs and can sustain conditional bargaining strategies; in incomplete-information models, battlefield outcomes and offers can also transmit information. Some specifications shift the parties' military position through combat, and their bargaining protocols leave at least one further offer or response after fighting. By contrast, the operation here neither changes the probability of victory nor reveals information on the selected equilibrium path, and the target has no move after its initial response. Spaniel and İdrisoğlu (2024) derive substitution toward cheaper and weaker options as the powerful option's cost rises. The substitution here runs across two histories at once, because the same operation can replace war after resistance and settlement after compliance.

A second line of work shows that a capability can hurt the state that holds it. Schram (2022) shows that a publicly observed improvement in a defender's hassling capability can reduce every defender type's payoff through predictability or emboldening. Kenkel and Schram (2025) show that a defender's private strength need not have the standard monotone relationship with war or its equilibrium payoff once a low-level response is available. A better gray-zone response can also push a resolute challenger toward war (Gannon et al. 2024). These models share a setup: the limited instrument belongs to the responder, the challenger moves first, and the analysis asks how the responder's type or capability changes the challenge. Our setup instead gives the instrument to an informed coercer that chooses its demand before the target responds and chooses its continuation action afterward. The target therefore adjusts its concession to how it expects the same actor to use the later menu, enabling three comparisons. First, we derive how the operation changes punishment after resistance and protection after compliance, which together determine coercive ability. Second, we show how the smaller concession can cause war among coercer types that never use the operation. Third, we compare the capability holder's payoff type by type, including types that settle or fight rather than operate. The operation need not reveal the coercer's type on the selected equilibrium path; the target responds to anticipated type-contingent actions.

The paper's third contribution is to study coercive ability explicitly as a composite of threat and assurance. Coercion is widely understood to require not only a credible threat but also credible assurance that compliance will end the harm (Cebul, Dafoe, and Monteiro 2021; Pauly 2024; 2025). Yet few formal models analyze both together. Pape (1996) classifies coercive strategies by how they make resistance costly or futile, emphasizing the threat. We add assurance—the harm that compliance prevents—and show that a cheaper or more capable operation can reduce concessions when its effect on assurance exceeds its effect on the threat. Existing accounts trace assurance failure to shifts in power (Powell 1996; 2006), inability to commit against future demands (Cebul, Dafoe, and Monteiro 2021; Haun 2015; Sechser 2018), loss of control over punishment (Pauly 2024), or reputational costs (Sechser 2010). Two formal models come closest. Coe and Vaynman (2020) show that monitoring must verify compliance without exposing the monitored state to exploitation, but their tension operates through an endogenous shift in power. Kydd and McManus (2017) let the coercer announce a goal, the target choose an allocation, and the coercer attack even when the allocation meets the goal; assurance can lower the coercer's bottom line and the target's allocation. Our framework instead holds power fixed, has the coercer choose the demand, and gives it the same menu of settlement, general war, and a subwar operation after either target response. This structure places threat and assurance in the same acceptance constraint: expected harm after resistance supports a concession, whereas expected harm after compliance reduces it. Their difference determines coercive ability. To our knowledge, this is the first formal analysis to study coercive ability through this composite structure, which lets us derive how a capability changes both components, identify war caused by the smaller concession, and compare its payoff effect across coercer types. In the security-dilemma literature, assurance is a signal that one state sends and the other must believe (Kydd 2000). Assurance here is an equilibrium outcome instead: the target infers it from the action the coercer will find optimal once the transfer is in hand.

The Model

The analysis compares two crisis-bargaining games. They share the same players, information, demand, and target response. They differ only in the actions available to the coercer after the target responds. This design identifies how adding the operation to the coercer's menu changes punishment after resistance, protection after compliance, and the demand the target accepts.

Figure 1 shows the common timing. The coercer first makes a take-it-or-leave-it demand. The target then concedes or resists. Finally, the coercer chooses a continuation action. The coercer's cost of war is private information, and abandoning a resisted demand carries an audience cost (Banks 1990; Fearon 1994; Schultz 1999). The benchmark continuation menu contains settlement and war. The second game adds a subwar operation after either target response.

The final stage determines both requirements of coercion. The action chosen after resistance determines the threat. The action chosen after concession determines how much protection compliance provides. A continuous demand measures the target's response to both. Unlike models in which the proposer screens an uncertain target, private information here determines the coercer's conduct after the target has moved (Fearon 1995; Powell 1996).

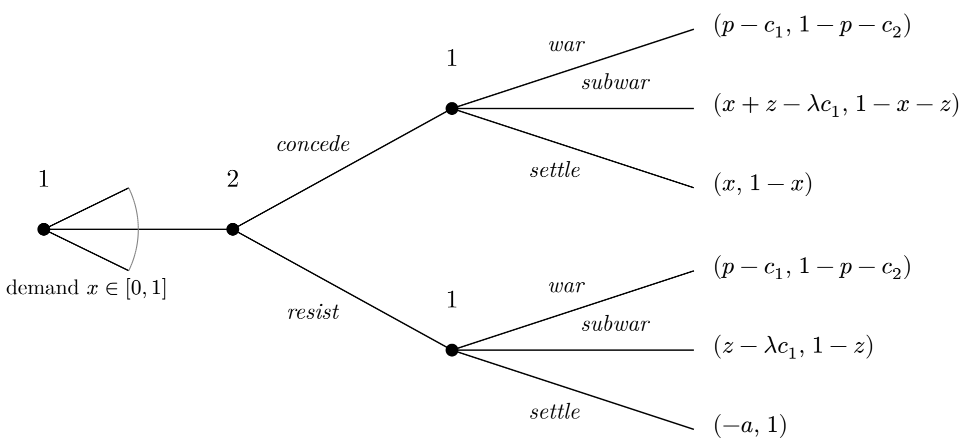

Figure 1: The Game

Note: Player 1 is the coercer, player 2 the target; the coercer's payoff appears first. The coercer demands x, and y denotes the transfer at the continuation stage: $y = x$ after concession, $y = 0$ after resistance. The benchmark deletes both operation branches. War reopens the dispute, so war payoffs do not depend on the transfer. The operation preserves the transfer, produces a bounded gain z, and costs λc_1. Settling after resistance costs the coercer a.

Players, Actions, and Timing

Two states value a disputed good at 1. The *target*, player 2, initially holds the good. The *coercer*, player 1, has a war cost c_1 drawn uniformly from $[0, \bar{c}]$. The coercer observes its cost, but the target knows only its distribution F, with density $f = \frac{1}{\bar{c}}$. A lower cost denotes a more resolute coercer. The target's war cost $c_2 \geq 0$ is common knowledge.

The coercer demands $x \in [0, 1]$. The target either concedes the demand or resists and transfers nothing. Let y be the transfer in place at the continuation stage. Thus $y = x$ after concession and $y = 0$ after resistance. The coercer chooses from the same continuation menu after either response.

Settlement leaves the transfer y in place. If the target resisted, settlement also imposes an audience cost $a > 0$ on the coercer for abandoning its demand. War gives payoffs $(p - c_1, 1 - p - c_2)$, where $p \in (0, 1)$ is the coercer's probability of victory. Because war reopens the whole dispute, these payoffs do not depend on any earlier transfer.

The second game adds a *subwar operation*. The baseline puts different operations on a common payoff scale. Let

$$S(y) = \min\{z, 1 - y\}$$

denote the resulting gain to the coercer and harm to the target. The operation costs the coercer λc_1, where $\lambda \in (0, 1)$, and produces payoffs $(y + S(y) - \lambda c_1, 1 - y - S(y))$. This reduced form places different operations on a common payoff scale. It preserves the earlier transfer and bounds the operation's effect by z.

The analysis maintains the following parameter region.

Assumption 1 (Parameter Restrictions): (i) The operation has limited reach,

$$p + c_2 + 2z \leq 1$$

and (ii) uncertainty about the coercer's war cost is sufficiently large,

$$\frac{2p + c_2}{1 - \lambda} < \overline{c}$$

A war costs the target $p + c_2$ relative to keeping the disputed good. Assumption 1(i) ensures that the operation has its full effect at every accepted demand, so $S(y) = z$ there.[1] Assumption 1(ii) implies $\overline{c} > 2p + c_2$ in the benchmark and directly imposes the stronger bound needed in the operation game. These inequalities make the target's payoff from concession decrease with the demand. Additional conditions governing interior cutoffs and the coercer's continuation menu are stated with the results that use them.

Comments on the Model

Model comparison. Following Paine and Tyson (2020), we adopt the experimental perspective on model comparison. Every primitive and move is fixed except the coercer's continuation menu, so differences between the games arise from the capability and the target's response to it. The comparison is nonetheless between a selected equilibrium of one game and a selected equilibrium of another, and the section on equilibrium selection states what that selection delivers and what it does not.

Reduced form. The model uses a common reduced form for different subwar operations rather than distinguishing their physical forms. The baseline sets the coercer's gain from the operation equal to the harm it imposes on the target. Corollary 1 separates these effects

[1]If war occurs with positive probability after compliance, the post-compliance war cutoff is positive, so $x^* < p - z$ when the operation is active and $x^* < p$ when it is dominated; either way $x^* + z < h + z \leq 1 - z$. If war does not occur after compliance, then $\Omega^* = x^*$, and (6) gives $x^* = \Theta^* - \Delta^* \leq \Theta^* \leq \max\{h, z\}$, so $x^* + z \leq \max\{h + z, 2z\} \leq 1 - \min\{z, h\}$. The benchmark is the case $z = 0$.

and shows how each enters the result. The analysis therefore treats z as a payoff effect, not as a description of what the operation physically does.

Payoff scope. The payoffs impose two scope conditions. First, general war supersedes an earlier transfer, as in standard bargaining models of war (Fearon 1995; Powell 1996). All main results are robust to allowing war to reclaim only part of an accepted concession; §F.2. Appropriative War gives the extension. Second, for tractability, the subwar operation preserves the transfer and does not trigger general war within the model's horizon. This condition makes post-compliance conduct independent of the size of the demand. After the target pays, settlement and the operation both preserve the transfer. The transfer cancels when the coercer compares those actions, so paying more cannot buy greater restraint.

Escalation risk. If an operation triggered general war with positive probability, using it would place the transfer at risk. The operation–settlement cutoff then falls in the transfer, so a larger concession raises the probability of settlement after compliance, which is the response the fixed-effect specification removes. A small escalation risk therefore weakens the mechanism, and the weakening grows with the risk. §F. Extensions and Scope states the extended cutoff and signs both comparative statics. The model applies to operations that occur while an accepted arrangement remains in force and treats later general war as outside its horizon.

Irreversible concessions. The model treats a completed transfer as irreversible within its horizon. This focus is deliberate: the assurance problem is most severe when the target cannot reverse its concession after further force. The assumption fits vacated territory, a destroyed capability, or a surrendered position. It fits staged or reversible concessions less well, including inspection access, an enrichment cap, or a force posture. Modeling staged implementation would require another target response after the operation and lies outside the baseline.

Capability possession. Possession is exogenous because the paper asks how having the capability changes bargaining, not why states acquire it, a question studied elsewhere

(Schram 2022). The model compares a coercer that has the capability with one that does not; it does not model investment, disclosure, divestment, or selection into acquisition. Proposition 6 therefore compares payoffs conditional on possession rather than providing an acquisition rule. The analysis also distinguishes bargaining performance from the coercer's payoff. Coercive ability measures the accepted-demand component retained without general war, whereas ex ante payoff also values the continuation menu. Different conditions govern these two quantities.

Operating costs. The operation costs λc_1. Its cost rises with the coercer's war cost but less rapidly because $\lambda < 1$. This specification represents operations that use the same underlying capabilities as war at a smaller scale. Type heterogeneity in operating cost is load-bearing: with a type-independent cost, all types rank the operation and settlement identically after compliance, so α is zero or one except at a knife edge. The operation carries no audience cost in the baseline. Only the visible abandonment of a resisted demand does. A public demand can create such a cost, while private threats can reduce it. Alternative bargaining protocols can also change crisis outcomes (Fey and Kenkel 2021; Kurizaki 2007). Both features remain fixed across the games; the extension below varies the operation's political cost explicitly.

Prior signals. The model omits signals sent before the demand, including public statements and military threats (Fearon 1994; 1997; Schultz 1999; Slantchev 2005). We omit these features because they would be present in both games. Holding prior signals fixed leaves the continuation menu as the only source of differences. This choice separates the mechanism from informational accounts of inexpensive force. Post (2019) finds that low-cost air mobilizations reveal little resolve and can reduce the success of compellent threats. The on-path demand conveys no information in the pooling equilibria studied here. The operation instead changes expected harm after both resistance and compliance.

The Benchmark

Begin with the benchmark by setting $z = 0$. The coercer can then choose only settlement or war after either target response. We compare the games using the amount obtained without reopening the dispute.

Definition 1 (Coercive Ability): A coercer's *coercive ability* Ω is the expected portion of an accepted demand that remains with the coercer without general war. If the target concedes x, then

$$\Omega = x\,[1 - \Pr(\text{war})]$$

where $\Pr(\text{war})$ is the probability of war after concession. War reopens the dispute and reverses the transfer. Throughout, *coercive ability* refers only to this paper-specific measure. It excludes any additional amount taken through the subwar operation and all continuation costs. Proposition 6 separately reports the coercer's expected total payoff.

This definition applies to both games. The solution concept is perfect Bayesian equilibrium. We focus on pooling equilibria and select the largest accepted demand in each game. The results are robust to allowing separating demands.[2] We retain only equilibria that satisfy the intuitive criterion (Cho and Kreps 1987), and the target concedes when indifferent.[3] The convention makes the binding on-path demand attainable and makes zero an accepted

[2]Lemma 5 in the Supporting Information establishes this robustness. Every type that does not fight after concession strictly prefers the largest accepted demand and therefore pools there. Only types that fight after both target responses can separate. Because those types give the target the same payoff after concession and resistance, their separation does not change the target's acceptance condition. The pooling equilibrium therefore attains the largest accepted demand and is coercer-optimal even when separating demands are allowed.

[3]D1 is stronger, and its verdict is symmetric across the two games. Proposition 8 in the Supporting Information shows that no equilibrium with pure target responses and an interior post-compliance war cutoff survives D1 in either game. The criterion therefore empties the class within which $x^\dagger$ and x^* are compared rather than selecting a different demand in one game. Resolute coercer types are good news for the target because concession moves them away from war, which inverts the usual logic of D1. If $c_2 > 0$,

demand when the target is indifferent. Under rejection at equality, the selected demand is the supremum of accepted demands and the reported outcomes are the corresponding limits. A dagger marks benchmark quantities. Unmarked symbols refer to the game with the operation. The Supporting Information contains all proofs.

At a fixed demand, the operation cannot mechanically reduce coercive ability. If it replaces war after compliance, the transfer survives more often. If it replaces settlement, the transfer remains in place. Any reduction in coercive ability must therefore arise because the capability changes the demand the target accepts.

To solve the benchmark, start at the continuation stage. War becomes less attractive as c_1 rises, while the settlement payoff is constant in c_1. After transfer y, the coercer fights when

$$c_1 \leq \kappa^\dagger(y) = p - y + a\mathbb{1}[y = 0]$$

and settles otherwise. A larger transfer lowers this cutoff because it makes settlement more valuable.

Concession weakly raises every coercer type's continuation value. A type that settles receives x after concession instead of paying a after resistance. A type that fights receives $p - c_1$ after either response and is indifferent. Thus no type prefers resistance, and every settling type prefers the largest accepted demand.

We therefore consider a pooling equilibrium in which the demand conveys no information about c_1. Fighting types could make other demands because they are indifferent, but these deviations do not change the target's loss from war. Under the conditions below, they do not change the largest accepted demand, the probability of war, or coercive ability.

The target's acceptance decision determines the pooling demand. After resistance, the coercer fights when $c_1 \leq p + a$. After concession of x, the cutoff falls to $p - x$ because settlement now preserves the transfer and carries no audience cost. Types in $(p - x, p +$

the elimination is independent of the tie rule; at $c_2 = 0$, it is tie-breaking-sensitive. We impose the intuitive criterion and do not claim that the pooling outcomes survive D1.

$a]$ therefore fight after resistance but settle after concession. Conceding costs the target x only when war does not reopen the dispute, while the benefit is the war loss avoided for these types. The target accepts when

$$x(1 - F(p - x)) \leq (p + c_2)(F(p + a) - F(p - x))$$

The largest accepted demand makes this inequality bind. Under the uniform distribution, the binding condition is a quadratic whose positive root is $x^\dagger$.

> Proposition 1 (Benchmark Equilibrium and Coercive Ability): Suppose the relevant war cutoffs are interior at every accepted demand. Under Assumption 1, a pooling equilibrium exists in which every type demands $x^\dagger$ and the coercer fights if $c_1 \leq \kappa^\dagger(x^\dagger)$. Among equilibria with pure target responses, this is the unique outcome at the largest accepted demand, and it satisfies the intuitive criterion. The coercer fights with probability $F(p + a)$ after resistance and $F\big(\kappa^\dagger(x^\dagger)\big)$ after concession. Its coercive ability is given by (2), and settlement after compliance occurs with probability $\alpha^\dagger = 1 - F\big(\kappa^\dagger(x^\dagger)\big)$.

The largest accepted demand is

$$x^\dagger = \frac{1}{2}\left[-(\bar{c} - 2p - c_2) + \sqrt{(\bar{c} - 2p - c_2)^2 + 4(p + c_2)a}\right] \tag{1}$$

Proposition 1 determines both the demand and continuation behavior. Types below the cutoff fight after concession, while types above it settle. Greater military power raises the cutoff and makes war optimal for a larger set of coercer types.

$$\Omega^\dagger = \underbrace{(p + c_2)\, F(p + a)}_{\text{punishment after resistance}} - \underbrace{(p + c_2)\, F\big(\kappa^\dagger(x^\dagger)\big)}_{\text{harm not prevented by compliance}} \tag{2}$$

Equation (2) writes coercive ability as one difference. The first term is the expected punishment produced by resistance. The second is the expected harm that compliance fails to prevent. A larger concession lowers the post-compliance war cutoff, makes settlement more likely, and reduces the second term.

The Game with a Subwar Operation

Now set $z > 0$ and solve the enlarged continuation menu. The payoffs from war, the operation, and settlement have slopes -1, $-\lambda$, and 0 in c_1. Because $0 < \lambda < 1$, the optimal action changes from war to the operation and then to settlement as the coercer's war cost rises. The relevant cutoffs are

$$\kappa_W(y) = \frac{p - y - S(y)}{1 - \lambda} \quad \text{and} \quad \kappa_L(y) = \frac{S(y) + a\mathbb{1}[y = 0]}{\lambda} \tag{3}$$

If $\kappa_W(y) < \kappa_L(y)$, low-cost types fight, intermediate types use the operation, and high-cost types settle. If the inequality is reversed, the operation is dominated and the coercer chooses between war and settlement.

Let $W(y)$ be the probability of war after transfer y, and let $L(y)$ be the probability of either war or the operation. When the operation is optimal for some types, $W(y) = F(\kappa_W(y))$ and $L(y) = F(\kappa_L(y))$. When the operation is dominated, define $\kappa_S(y) = p - y + a\mathbb{1}[y = 0]$ and set $W(y) = L(y) = F(\kappa_S(y))$. The same notation therefore covers both continuation menus.

Before solving for the demand, hold transfer y fixed and compare continuation behavior. This exercise does not assume that the equilibrium demands are equal. Whenever the operation is active, $\kappa_W(y) < \kappa_S(y) < \kappa_L(y)$. Types between the first two cutoffs switch from war to the operation. Types between the last two switch from settlement to the operation. Replacing war reduces the target's loss when war is more damaging than the operation. Replacing settlement increases the target's loss by $S(y)$. These substitutions change the threat after resistance and assurance after compliance.

The cutoffs determine when the operation enters the coercer's menu. When $S(y) = z$, it is optimal for some types after resistance if and only if $z + a > \lambda(p + a)$, and after concession of x if and only if $z > \lambda(p - x)$. Greater reach or lower relative cost makes the operation more attractive after either response. A larger audience cost also favors the operation after resistance because using it avoids the cost of backing down. A larger concession favors the operation over war after compliance because the operation preserves the transfer and war does not.

Having solved the coercer's continuation choice, turn to the target's response. The target concedes x when its expected loss after resistance is at least as large as its expected loss after compliance:

$$\begin{aligned} &(p + c_2)W(0) + z(L(0) - W(0)) \\ &\quad \geq x(1 - W(x)) + (p + c_2)W(x) + S(x)(L(x) - W(x)) \end{aligned} \tag{4}$$

The left side of (4) is expected punishment after resistance. On the right, the first term is the transfer that survives unless war reopens the dispute. The remaining terms are harm that compliance does not prevent. The target's acceptance decision therefore depends on continuation behavior after both responses. The equilibrium demand is the largest x that satisfies this inequality.

Let α be the probability of settlement after compliance in the pooling equilibrium. A closed-form solution requires four additional conditions. The war cutoffs and the post-compliance operation–settlement cutoff are interior. The operation is active after compliance. Every nonfighting type uses the operation after resistance, so $\kappa_L(0) \geq \overline{c}$ or $z + a \geq \lambda\overline{c}$. Finally, $\alpha > 0$.

Each condition has a separate role. Interior cutoffs make action probabilities linear under the uniform distribution. Activity after compliance ensures that all three continuation actions matter. The condition on $\kappa_L(0)$ sets $L(0) = 1$, so every type that does not fight after resistance uses the operation. A positive α leaves some types that use the operation after resistance willing to settle after compliance. Compliance protects the target from the operation for exactly these types. Assumption 1(i) keeps $S(y) = z$ at accepted demands. These restrictions produce the closed form below. The substitution argument above does not require them.

Proposition 2 (Equilibrium with a Subwar Operation): Under Assumption 1 and the additional conditions above, a pooling equilibrium exists in which every type demands x^* and war occurs with probability $W(x^*)$. Among equilibria with pure target responses, this is the unique outcome at the largest accepted demand. This pooling equilibrium satisfies the intuitive criterion.

When $\alpha > 0$, the largest accepted demand is

$$x^* = \frac{1}{2}\Big[-((1-\lambda)\bar{c} - 2p - c_2 + 2z) + \sqrt{((1-\lambda)\bar{c} - 2p - c_2 + 2z)^2 + 4(1-\lambda)\bar{c}z\alpha}\Big] \tag{5}$$

The constant term inside the quadratic for x^* is proportional to $z\alpha$. The term is positive only when compliance moves a positive mass of coercer types from the operation to settlement. If $\alpha = 0$, every type uses force after compliance, so compliance provides no protection from the operation and the target accepts no positive demand. The next result determines α and explains why the accepted transfer may fail to increase it.

Consider the operation–settlement choice after compliance. In the benchmark, a larger transfer raises the settlement payoff but leaves the war payoff unchanged. It therefore makes settlement more likely. With the operation, both settlement and the operation preserve the transfer. Their payoff difference is $S(y) - \lambda c_1 - C(y)$, where $C(y)$ is any additional operating cost. A larger transfer changes this comparison only if it changes the operation's net return $S(y) - C(y)$.

Proposition 3 (Settlement After Compliance): Suppose the operation takes $S(y)$ and costs $\lambda c_1 + C(y)$ after transfer y. At any accepted demand at which the operation is active and the operation–settlement cutoff is interior, $\kappa_L(y) = \frac{S(y)-C(y)}{\lambda}$ and its derivative is $\frac{S'(y)-C'(y)}{\lambda}$. In this model, $S(y) = z$ and $C(y) = 0$

at every accepted demand, so $\kappa_L(y) = \frac{z}{\lambda}$ is constant in the demand and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$. Settlement becomes less likely as z rises and more likely as λ rises.

Proposition 3 characterizes one continuation boundary at an accepted demand, not the full acceptance decision. Under the fixed-effect specification, a larger transfer can move types from war to the operation through κ_W. It cannot move them from the operation to settlement because κ_L is constant. With a general benefit or cost technology, $S'(y) - C'(y)$ determines that second response. The target still chooses the demand according to (4). In Figure 3, the constant cutoff appears as the horizontal upper boundary of the region where the operation replaces settlement.

Spaniel (2023) studies the neighboring case in which a limit on what can be taken is itself pacifying, and finds a non-monotonic relation: war is least likely at moderate indivisibility, because an extreme one leaves no acceptable agreement (his Propositions 1 and 2). His limit is on the divisibility of the stake rather than on the instrument. Proposition 5 below is the mirror image of the same comparison: removing a binding restriction on what the coercer may take after compliance can leave every coercer type weakly worse off.

In the baseline, z is at once the coercer's gain and the target's harm. Separating the two identifies which primitive the assurance channel runs on.

Corollary 1 (Gain and Harm): Suppose the operation imposes $d > 0$ on the target and delivers $g \geq 0$ to the coercer, with the two unlinked. Then $\kappa_L(y) = \frac{g + a\mathbb{1}[y=0]}{\lambda}$ and $\alpha = 1 - F\left(\frac{g}{\lambda}\right)$, so settlement after compliance depends on g and not on d, and the corner at which coercive ability is zero becomes $g \geq \lambda\bar{c}$. If $g = 0$, then $\alpha = 1$, the post-compliance node is the benchmark's, and that corner is unreachable. With interior post-resistance cutoffs and an active punishment operation, $a(1 - \lambda) > \lambda p$, the operation raises the accepted demand and coercive ability if and only if $d > \lambda h$. The present model sets $d = g = z$.

The target's loss d governs how much harm each operation does. The coercer's gain g governs whether compliance protects at all. An operation that provides no benefit still costs λc_1 to conduct, so no type conducts it once the transfer is in hand. The assurance channel therefore depends on the operation's benefit to the coercer, not on target harm alone.

Now return to resistance and ask whether the operation replaces war. If the operation is active after resistance, war occurs with probability $F\left(\frac{p-z}{1-\lambda}\right)$; otherwise it occurs with probability $F(p+a)$. The operation is active, and lowers the probability of war, if and only if $z+a>\lambda(p+a)$. It never raises war after resistance.

The baseline assigns no audience cost to the operation, so it provides a way to avoid backing down after resistance. Suppose instead that using the operation imposes an audience cost $a_L \in [0,a]$ after either response. Two boundaries then govern the extension. The operation replaces war after resistance if and only if $z+a-a_L>\lambda(p+a)$, and it remains in the post-compliance menu if and only if $a_L<z-\lambda(p-x)$; past that boundary compliance fully protects against the operation, although war may still follow. §F.3. The Operation's Audience Cost states the cutoffs and the modified acceptance condition. The two political costs have distinct roles: a makes the operation relatively attractive after resistance, while a_L makes it less attractive after both responses. The comparison is therefore conditional on their joint configuration, not on publicity alone.

The probabilities π and α describe only two continuation actions. The resistance-stage comparison holds the history fixed and does not determine the equilibrium probability of war after the target complies, because the capability can change the accepted demand. To evaluate the target's full decision, convert every continuation outcome into expected loss. Let Θ^* denote punishment after resistance and Δ^* denote harm not prevented by compliance.

Remark 1 (Coercive Ability as Threat Minus Assurance Failure): If a common accepted demand leaves the target indifferent, coercive ability equals expected punishment after resistance, Θ^*, minus harm not prevented by compliance, Δ^*. It is zero exactly when $\Theta^* = \Delta^*$. This identity follows from (4) and does not require the closed-form equilibrium.

$$\begin{aligned}\Omega^* = &\underbrace{(p+c_2)W(0) + z(L(0) - W(0))}_{\text{threat: punishment after resistance } =:\Theta^*} \\ &- \underbrace{(p+c_2)W(x^*) + S(x^*)(L(x^*) - W(x^*))}_{\text{assurance failure: harm not prevented by compliance } =:\Delta^*}\end{aligned} \tag{6}$$

At the largest common accepted demand, the target is indifferent. Rearranging (4) gives $x^*(1 - W(x^*)) = \Theta^* - \Delta^*$. Let $\Theta^\dagger$ and $\Delta^\dagger$ be the corresponding benchmark quantities from Proposition 1 and (2). The difference between the games is

$$\Omega^* - \Omega^\dagger = \left(\Theta^* - \Theta^\dagger\right) - \left(\Delta^* - \Delta^\dagger\right) \tag{7}$$

The operation increases coercive ability when its effect on punishment exceeds its effect on post-compliance harm. It reduces coercive ability when the reverse inequality holds. The operation weakens the threat when it lowers Θ^* and weakens assurance when it raises Δ^*. Which effect occurs depends on whether the operation replaces war or settlement after each target response. Setting $z = 0$ removes the operation terms and recovers (2). The following sections compare these two changes.

Threat, Assurance, and Coercive Ability

The operation's effect depends on the action it replaces after each target response. Replacing settlement after resistance increases punishment and strengthens the threat. Replacing a more damaging war after resistance reduces punishment and weakens the threat. After compliance, replacing settlement adds harm and weakens assurance, while replacing a more damaging war removes harm and strengthens assurance. Different coercer types can make different substitutions, so the four effects may reinforce or offset one another.

The shaded regimes include the target's endogenous demand response. To see the underlying continuation choices, first hold the transfer fixed. Adding the operation can change behavior in only two ways. Some coercer types use it instead of war, reducing the target's loss from h to z. Others use it instead of settlement, increasing the target's loss from zero to z. The operation's net effect at any history is the harm added by former settlers minus the harm removed by former fighters.

With interior cutoffs and a uniform cost distribution, these two groups have a simple relation: for every type shifted out of war, $\frac{1-\lambda}{\lambda}$ types are shifted out of settlement. Expected harm therefore rises when $z > \lambda h$, because the harm added by former settlers dominates, and falls when $z < \lambda h$, because replacing war dominates. The equality $z = \lambda h$ marks the balance between the two groups. §D.9. Interior Switch Accounting gives the cutoff algebra.

The same logic applies after resistance and compliance, but the number of types switching at each history differs. The audience cost makes the operation more attractive after resistance because it avoids the penalty for backing down. The accepted transfer makes it more attractive after compliance because the coercer keeps what the target has already paid. When the operation adds harm, switching after resistance strengthens coercion and switching after compliance weakens it. When the operation replaces a more damaging war, these effects reverse. Coercive ability depends on which switch is larger.

Two equilibrium responses qualify this simple comparison. One switching group may be exhausted. In the closed-form region, every type that avoids war after resistance already uses the operation, whereas some types still settle after compliance. The target can also respond by accepting a smaller demand, which may produce more war after compliance. Together, these responses allow the operation to weaken both threat and assurance. §D.9. Interior Switch Accounting gives the exact conditions.

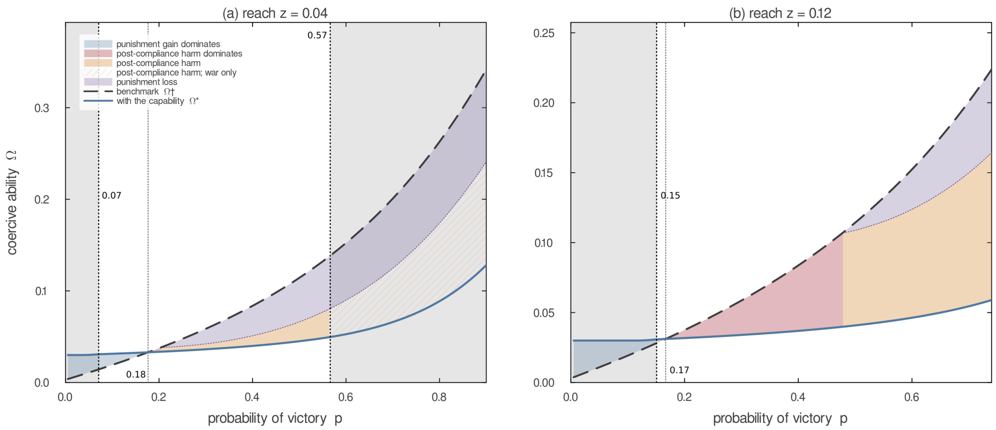

Figure 2: Assurance Loss Can Reduce Coercive Ability Despite Greater Punishment

Note: Coercive ability in the benchmark (dashed) and with the capability (solid) as the coercer's probability of victory varies. Blue marks where stronger punishment dominates added post-compliance harm; red marks where post-compliance harm dominates despite stronger punishment; orange marks where both channels reduce coercive ability. Hatching indicates that the operation is not used after compliance, so the remaining harm comes from war induced by the smaller demand. Grey lies outside the closed-form region. Parameters: $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, c_1 uniform on $[0, 2]$; $z = 0.04$ in panel (a) and $z = 0.12$ in panel (b).

Equation (7) remains the general comparison. Coercive ability rises when punishment increases more than post-compliance harm and falls when the reverse is true. The accepted demand, the probability of war, and coercive ability consequently need not move together.

Figure 2 illustrates this comparison at two values of the operation's reach. In both panels, greater punishment after resistance initially outweighs added post-compliance harm. As power rises, the coercive-ability curves cross before the punishment comparison does: post-compliance harm first becomes the dominant loss, and only later do both channels reduce coercive ability. The figure visualizes possible substitutions; the analytical conditions, not the plotted slices, carry the result.

The two panels also differ in what produces the loss. At the smaller reach, no coercer type eventually chooses the operation after compliance; beyond that point the loss comes from weaker punishment after resistance together with the war that the smaller demand no longer prevents. At the larger reach, post-compliance harm combines the operation's direct effect with additional war from the smaller demand. Proposition 4 gives the exact comparison. The panels illustrate its channels but do not establish their prevalence.

The fixed-effect specification makes the assurance response especially clear. After compliance, a larger transfer raises the payoffs from settlement and the operation by the same amount. It leaves the operation–settlement cutoff unchanged, although it can still move types from war to the operation. Proposition 3 shows that a general benefit or cost technology restores a response through $S'(y) - C'(y)$.

Both equilibrium outcomes admit closed forms, allowing an exact comparison of coercive ability.

Proposition 4 (Coercive Ability in Closed Form): If the benchmark war cutoffs are interior at every accepted demand, $\Omega^\dagger = \frac{h(x^\dagger + a)}{\overline{c}}$. Under the conditions for (5), $\Omega^* = \frac{x^*(h-z)}{(1-\lambda)\overline{c}} + z\alpha$. The capability lowers coercive ability if and only if:

$$\frac{h(x^\dagger + a)}{\overline{c}} > \frac{x^*(h-z)}{(1-\lambda)\overline{c}} + z\alpha \tag{8}$$

Because (1) and (5) express both demands in primitives, (8) is an exact condition. In the benchmark, coercive ability equals the target's loss from war, h, multiplied by the sum of the demand and audience cost and divided by the support of the coercer's war cost. The audience cost enters because it makes abandoning a resisted demand less attractive.

With the operation, the two terms in Ω^* correspond to two groups whose behavior differs across the target's responses. The probability of war falls by $\frac{x^*}{(1-\lambda)\overline{c}}$ between resistance and compliance. For the types responsible for this difference, compliance replaces war, which costs the target h, with the operation, which costs z. Their contribution to coercive ability is therefore $\frac{x^*(h-z)}{(1-\lambda)\overline{c}}$. A second group uses the operation after resistance but settles after compliance. This group has mass α, and compliance protects the target from harm z, producing the term $z\alpha$.

The second term is single-peaked in reach because $z\alpha = z\left(1 - \frac{z}{\lambda \bar{c}}\right)$, which is maximized at $z = \frac{\lambda \bar{c}}{2}$. Greater reach increases the harm avoided for every type in the second group, but it also shrinks that group by making the operation preferable to settlement for more types after compliance. The two terms may move in opposite directions, so neither reach nor the probability of war alone signs the comparison. The resistance-stage condition determines when war after resistance falls, but the target also accounts for operations that replace settlement.

> Remark 2 (Power): In the closed-form region, the protection term $z\alpha$ depends on reach, relative cost, and the spread of war costs but not on the probability of victory. Total Ω^* still depends on power through both x^* and $h - z$, so the derivative comparison with $\Omega^\dagger$ is not signed in general.

The same cutoff identifies complete coercive failure.

> Remark 3 (Complete Coercive Failure): Coercive ability is zero if and only if $z \geq \lambda \bar{c}$. When the inequality holds, the operation–settlement cutoff after compliance is at or above the top of the type distribution, so even the highest-cost coercer type prefers the operation to settlement after compliance. Compliance then provides no protection from the operation, $\alpha = 0$, the target refuses every positive demand, and $x^* = \Omega^* = 0$.

The expression $\alpha = 1 - \frac{z}{\lambda \bar{c}}$ applies only for $z \leq \lambda \bar{c}$; above that value the cutoff leaves the support of the war-cost distribution and $\alpha = 0$. Remark 3 itself does not depend on the closed-form quadratic. Its proof uses only the target's gain from conceding an arbitrarily small demand and the monotonicity of the acceptance margin, both established in the Supporting Information: when $\alpha = 0$ that gain is zero, and the margin is strictly decreasing at every larger demand.

Figure 3: The Operation Substitutes for War and Settlement and Can Cause War

Note: Panel (a) colors each combination of the coercer's probability of victory and war cost by the change in the post-compliance outcome when the operation becomes available, at the calibration of Figure 2 with $z = 0.04$. The operation region's upper boundary is horizontal because $\kappa_L = \frac{z}{\lambda}$ contains neither p nor the demand. Panel (b) plots the probability of war after compliance on each game's equilibrium path: below $p \approx 0.29$ the operation replaces war; above it, the smaller accepted demand pushes types that would have kept the benchmark settlement into war. Hatching marks the range above $p \approx 0.57$, where the operation is dominated after compliance. There the post-compliance menu is the benchmark menu evaluated at a smaller demand, and the entire increase follows from the weaker punishment the target expects after resistance.

How Peace Fails

The preceding results determine how much the target concedes. They also change the continuation outcome. Figure 3 compares post-compliance actions across the two games for each combination of power and war cost. Among low-cost types, the operation can replace war. Among intermediate types below κ_L, it replaces settlement. A third change arises from the target's response. When the capability lowers the accepted demand, types near the benchmark war cutoff receive too little to settle and instead choose war. These types never use the operation. The capability changes their behavior only because it reduces the concession.

The direct menu effect and the target's response move war in opposite directions. At a fixed concession x, an active operation lowers the war boundary from $p - x$ to $\kappa_W(x)$ and replaces war for some low-cost types. Equilibrium adds a reinforcing response through the concession. When the operation weakens punishment after resistance, holding out becomes less costly and the target concedes less. The smaller concession makes war more attractive after compliance. The resulting increase in post-compliance war weakens assurance, which further reduces what the target will concede. The equilibrium demand x^* incorporates this feedback. Algebraically, the relevant cutoffs are $\kappa^\dagger(x^\dagger) = p - x^\dagger$ and $\kappa_W(x^*) = \frac{p-x^*-z}{1-\lambda}$. War rises when the concession response moves the second cutoff above the first, causing benchmark settlers to fight.

Panel (b) plots the benchmark as the dashed black curve and the game with the operation as the solid red curve. To the left of their crossing at $p \approx 0.29$, the red curve lies below the benchmark: the pale gap shows the reduction in war as the operation replaces it. To the right of the crossing, the red curve lies above the benchmark: the shaded gap shows the additional war caused by the smaller concession. Between the crossing and the dotted line at $p \approx 0.57$, some coercer types still use the operation after compliance. In the hatched region to the right of the dotted line, the operation is dominated after compliance, so war rises even though the operation is never used on the equilibrium path. Additional war therefore means that more types choose war than in the benchmark, not that an operation occurs before war.

The unhatched and hatched ranges show two versions of this feedback. Between the crossing and $p \approx 0.57$, the operation remains available after compliance. Anticipated operation weakens assurance directly, so the target concedes less. The smaller concession then sends marginal types to war. To the right of $p \approx 0.57$, no type uses the operation after compliance. Instead, the operation replaces war after resistance and weakens the threat. The target responds by conceding less. That smaller concession sends benchmark settlers to war, so compliance prevents less harm and assurance weakens. This assurance loss reinforces the target's unwillingness to concede. These effects are jointly determined in equilibrium, not a temporal sequence. In both ranges, the types drawn into war never use the operation.

The Settlement Forgone

We next compare what the coercer obtains in the two equilibria. The relevant question is whether the operation's direct gain compensates for the smaller accepted demand.

Proposition 5 (The Settlement Forgone Exceeds the Operation's Gain): Suppose the benchmark post-compliance war cutoff is interior and the operation-game closed-form conditions hold. The benchmark demand exceeds the operation-game demand plus the operation's full direct gain, $x^\dagger > x^* + z$, if and only if (9) holds. Under the same condition, every coercer type is weakly worse off with the operation, a positive measure is strictly worse off, and war after compliance is strictly more likely with the operation than in the benchmark.

$$\big(x^\dagger - z\big)\big(1 - W\big(x^\dagger - z\big)\big) > \frac{\big(x^\dagger - z\big)(h - z)}{(1 - \lambda)\bar{c}} + z\alpha \tag{9}$$

Condition (9) evaluates the operation-game target's acceptance rule at the demand $x^\dagger - z$, whose concession would leave the coercer the benchmark settlement once the operation's full gain is added. The left side is the transfer that concession would surrender. The right side is the harm it would avoid: the term from types that compliance moves from war to the operation, and the protection term $z\alpha$ from Proposition 4. The condition states that the surrendered transfer exceeds the avoided harm, so the target refuses this demand and every larger one. Expanding W gives the same condition in primitives, recorded with the proof in the Supporting Information. In the limit of vanishing reach it reduces to $(p + c_2)a > \lambda\bar{c}\,x^\dagger$: a larger audience cost relaxes it and a larger relative cost of use tightens it.

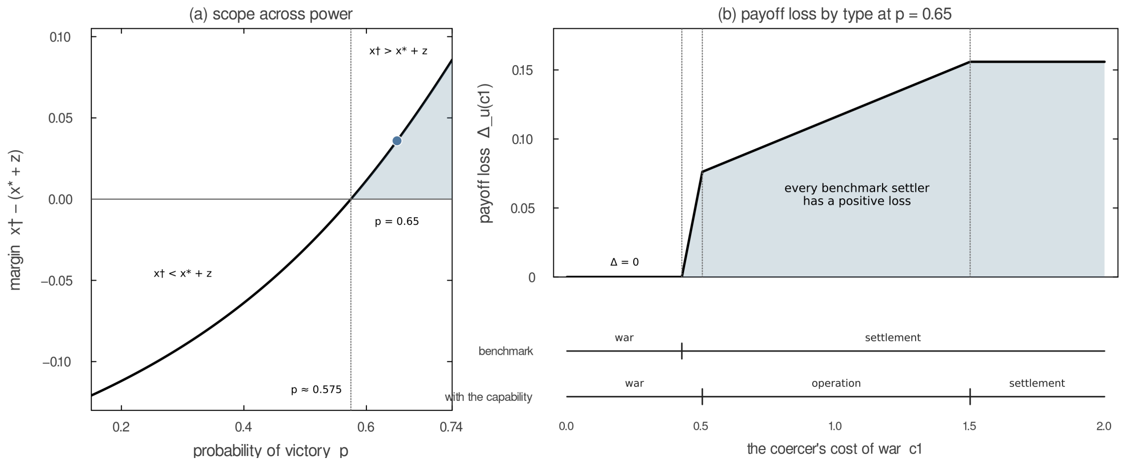

Figure 4: When the Forgone Settlement Exceeds the Operation's Gain

Note: Panel (a) plots $m(p) = x^{\dagger}(p) - x^{*}(p) - z$ over $p \in (0.15, 0.74]$, the interval where Proposition 5′s other premises hold at the displayed primitives; the margin crosses zero at $p \approx 0.57$, and the dot marks $p = 0.65$, where $m(p) = 0.04$. Panel (b) fixes $p = 0.65$ and plots $\Delta_u(c_1) = u^{\dagger}(c_1) - u^{*}(c_1)$, the benchmark payoff minus the operation-game payoff, with each game's continuation actions marked as intervals over the coercer's war cost: types through $c_1 = 0.43$ fight in both games and lose nothing; every higher type settles in the benchmark and strictly loses, with war through $c_1 = 0.50$, the operation through $c_1 = 1.5$, and settlement thereafter. Parameters: $c_2 = 0.02$, $a = 0.30$, $\lambda = 0.08$, $z = 0.12$, c_1 uniform on $[0, 2]$.

Define $m(p) = x^{\dagger}(p) - x^{*}(p) - z$. Figure 4 plots this margin over the values of p for which the proposition's other premises hold at the displayed primitives, then shows which coercer types bear the payoff loss at $p = 0.65$.

For the displayed parameter slice, Proposition 5′s condition holds throughout $p \in (0.57, 0.74]$. The comparison uses gross holdings: subtracting the operating cost only widens the payoff loss.

Panel (b) shows that the loss is not confined to types that conduct the operation. Types below $\kappa^{\dagger}(x^{\dagger})$ fight in both games and receive the same war payoff. Types between $\kappa^{\dagger}(x^{\dagger})$ and $\kappa_W(x^{*})$ settle in the benchmark but fight after the smaller concession. Types between $\kappa_W(x^{*})$ and $\kappa_L(x^{*})$ conduct the operation, while higher-cost types settle in both games at the lower demand x^{*}. Under Proposition 5′s condition, every benchmark settler strictly loses, even though only a subset uses the operation.

That the first of these groups exists is not a feature of the illustration. Proposition 5 implies $\kappa_W(x^*) > \kappa^\dagger(x^\dagger)$, so the same parameter region in which every coercer type is weakly worse off is also a region in which equilibrium war is strictly more likely. This does not contradict the resistance-stage comparison, because resistance fixes the transfer at zero and does not occur on the equilibrium path of either game.

The next result separates the value of the larger continuation menu from the cost imposed by the target's response.

> Proposition 6 (The Coercer's Expected Payoff): If the relevant cutoffs are interior and the operation is active after concession, the coercer's ex ante payoffs are given by (10). Each payoff equals the accepted demand plus the value of the continuation menu. The coercer is worse off with the operation if and only if $V^\dagger > V^*$.

$$\begin{aligned} V^\dagger &= x^\dagger + \frac{\left(p - x^\dagger\right)^2}{2\overline{c}} \\ V^* &= x^* + \frac{\left(p - x^* - z\right)^2}{2(1-\lambda)\overline{c}} + \frac{z^2}{2\lambda\overline{c}} \end{aligned} \tag{10}$$

To decompose this comparison, remain in the active, interior region and hold an accepted transfer y fixed. Let $Q_0(y)$ and $Q_1(y)$ be the coercer's expected continuation payoffs in the two games:

$$\begin{aligned} Q_0(y) &= y + \frac{(p-y)^2}{2\overline{c}} \\ Q_1(y) &= y + \frac{(p-y-z)^2}{2(1-\lambda)\overline{c}} + \frac{z^2}{2\lambda\overline{c}} \end{aligned}$$

Then $V^\dagger = Q_0\left(x^\dagger\right)$, $V^* = Q_1(x^*)$, and

$$V^* - V^\dagger = \underbrace{Q_1(x^*) - Q_0(x^*)}_{\text{menu at the same transfer}} + \underbrace{Q_0(x^*) - Q_0\left(x^\dagger\right)}_{\text{endogenous-demand response}}$$

The first term values the larger menu at the operation-game demand and equals $\frac{[\lambda(p-x^*)-z]^2}{2\lambda(1-\lambda)\bar{c}} \geq 0$. This menu gain is nonnegative because each type can retain its benchmark action. The second term holds the benchmark menu fixed and changes the transfer from $x^\dagger$ to x^*. It is negative whenever $x^* < x^\dagger$. The coercer's expected payoff falls exactly when the loss from the smaller accepted demand exceeds the value of the added action.

At the continuation stage, every type below $\frac{z}{\lambda}$ prefers the operation to settlement, although the lowest-cost types may prefer war. Every chosen action is optimal at the accepted demand. The loss in Proposition 6 arises before that stage, when the target responds to the menu by accepting less.

Scope of the Equilibrium Comparison

Part of the analysis does not depend on equilibrium selection. If the operation takes a fixed amount and is active after an accepted demand, the coercer settles if and only if $c_1 > \frac{z}{\lambda}$. That cutoff does not depend on the demand, the target's mixing probability, or off-path beliefs, and Proposition 7 in the Supporting Information proves it for every equilibrium. The threat-minus-assurance identity in Remark 1 is free of the closed form in the same way: it requires only a common accepted demand that leaves the target indifferent. Both claims are conditional in one respect. They apply once a demand at which the operation is active has been conceded. Different equilibria can place different demands on the path, change the types associated with each demand, or include accepted demands at which the operation is dominated, so the unconditional settlement rate and coercive ability are not fixed by the cutoff alone.

Both pooling equilibria satisfy the intuitive criterion. That restriction removes a type from an unexpected demand only when no sequentially rational target response lets the type match its equilibrium payoff. Allowing separating demands does not alter the selected outcome: Lemma 5 in the Supporting Information shows that only types that fight after both target responses can separate, and their demands do not affect the target's acceptance condition.

What the comparison selects in each game is the coercer-optimal equilibrium within the class analyzed. The Supporting Information establishes that every type's equilibrium payoff is its optimized continuation payoff at the largest accepted demand, which increases in that demand. The selection is not innocuous. A worst-case selection would make the comparison vacuous, because each game also has an equilibrium satisfying the intuitive criterion in which the target refuses every positive demand and $\Omega^\dagger = \Omega^* = 0$.[4] The comparison is deliberately best case against best case.

Conclusion

An additional military capability improves coercion only when it increases punishment after resistance more than it increases harm after compliance. A subwar operation can therefore help weak coercers whose threat of war is not credible and hurt strong coercers that already obtain favorable settlements. It replaces war for some coercer types and settlement for others, and because it also lowers the demand the target accepts, it can induce war among types that the benchmark settlement would have pacified. When the operation cannot compensate for the smaller concession, every coercer type is weakly worse off despite choosing its optimal continuation action. That is also the region in which war after compliance is strictly more likely than in the benchmark, so the parameters that are worst for the coercer are worst for peace as well.

These results also clarify why a capability's value in deterrence need not carry over to compellence. Both require punishment to be conditional, but the compellence problem here asks the target to make a concession and then rely on the coercer not to use the operation. Because the operation remains available after compliance, it can strengthen punishment after resistance while weakening the protection that compliance buys. A low-cost option

[4]Fix a belief concentrated on coercer types that take the same continuation action after either target response. Conceding x then leaves the target exactly x worse off than resisting, so refusal is strictly optimal at every positive demand, and no type is equilibrium dominated by the deviation. These are the types that settle after either response in the benchmark, which exist whenever $p + a < \overline{c}$, and the types that use the operation after either response in the operation game, which exist whenever $z > \lambda p$.

may therefore support a deterrent threat yet weaken a compellent bargain. The effect of a capability on compellence cannot be inferred from threat credibility alone (Cebul, Dafoe, and Monteiro 2021; Kydd and McManus 2017; Schelling 1966).

The analysis also qualifies the usual option-value logic. At a fixed transfer, adding the operation cannot hurt the coercer because every benchmark action remains available. Once the target chooses the transfer in anticipation, however, the larger menu can make the coercer worse off. The comparison identifies what a useful commitment would have to do: preserve punishment after resistance while limiting the operation after compliance. A coercer can gain from restricting an action it would rationally choose later because the restriction changes the bargain offered earlier. This gives a specific form to the power to bind oneself (Schelling 1960).

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A. Notation

Let $y = x$ after concession and $y = 0$ after resistance. A subwar operation produces a gain $S(y) = \min\{z, 1 - y\}$ and costs the coercer λc_1, so $S(0) = z$. Under Assumption 1(i), $S(y) = z$ at every relevant accepted demand. Settlement after resistance costs the coercer a, and $h = p + c_2$ is the target's loss from war. As in the main text, $W(y)$ is the probability of war after transfer y, while $L(y)$ is the probability of either war or the operation. When the operation is active, $W(y) = F(\kappa_W(y))$ and $L(y) = F(\kappa_L(y))$. When it is dominated, $W(y) = L(y) = F(\kappa_S(y))$. Finally, $\alpha = 1 - L(x^*)$ is the probability of settlement after compliance and $\pi = W(0)$ is the probability of war after resistance.

B. Proofs of the Lemmas

Lemma 1 (The Coercer's Menu in the Benchmark): Given transfer y, the coercer fights if and only if $c_1 \leq \kappa^\dagger(y) = p - y + a\mathbb{1}[y = 0]$ and settles otherwise.

Lemma 2 (No Type Prefers Resistance): For every c_1 and $x > 0$, the coercer's continuation value is weakly higher after concession than after resistance and strictly higher for types that do not fight after concession. If the operation is active after concession, the indifferent types are $[0, \kappa_W(x)]$, which has positive measure if and only if $p > x + S(x)$. If the operation is dominated, the indifferent types are $[0, \kappa_S(x)]$.

Lemma 3 (The Coercer's Menu with a Subwar Operation): If $\kappa_W(y) < \kappa_L(y)$, the coercer fights below $\kappa_W(y)$, uses the operation between the cutoffs, and settles above $\kappa_L(y)$. If the cutoffs are reversed, the operation is dominated. Equation (3) gives both cutoffs.

Lemma 4 (The Target's Rule): The target concedes x if and only if (4) holds.

B.1. Proof of Lemma 1

Proof. After transfer y, settlement pays $y - a\mathbb{1}[y = 0]$ and war pays $p - c_1$. The coercer fights when $p - c_1 \geq y - a\mathbb{1}[y = 0]$. Rearranging this inequality gives the cutoff in the lemma. $\square$

B.2. Proof of Lemma 2

Proof. Fix c_1 and $x > 0$. Settlement pays x after concession and $-a$ after resistance. War pays $p - c_1$ after either response. The operation pays $x + S(x) - \lambda c_1$ after concession and $z - \lambda c_1$ after resistance. If $z \leq 1 - x$, then $x + S(x) = x + z > z$. If $z > 1 - x$, then $x + S(x) = 1 > z$. Thus each continuation action pays at least as much after concession as after resistance.

War is the only action whose payoff does not strictly rise after concession. Because the war cutoff weakly decreases with y, a type that fights after concession also fights after resistance. That type receives $p - c_1$ after either response and is indifferent.

Now take a type that does not fight after concession. Because war was available but not chosen, its continuation value after concession exceeds $p - c_1$. Settlement and the operation also pay strictly more after concession because $x > -a$ and $x + S(x) > z$. The type's continuation value after resistance is the maximum of $p - c_1$, $-a$, and $z - \lambda c_1$. Each candidate is strictly below its corresponding post-concession payoff or below the post-concession continuation value. The type therefore strictly prefers concession.

If the operation is active, the indifferent set is exactly $[0, \kappa_W(x)]$ and has positive measure if and only if $p > x + S(x)$. If the operation is dominated, the indifferent set is $[0, \kappa_S(x)]$. These types' payoffs do not determine their demands. The pooling equilibrium assigns them x^*.

No type gains by inducing resistance. Settling types prefer the largest accepted demand. Operation users weakly prefer a larger demand and strictly prefer it under Proposition 2's condition that $S(y) = z$. Fighting types are indifferent. The pooling equilibrium assigns the same demand to every type, so the target evaluates that demand under the prior F. $\square$

B.3. Proof of Lemma 3

Proof. Fix transfer y. The coercer's three payoffs are

$$u_S = y - a\mathbb{1}[y=0], \quad u_W = p - c_1, \quad u_L = y + S(y) - \lambda c_1$$

Their slopes in c_1 are 0, -1, and $-\lambda$. Since $-1 < -\lambda < 0$, the optimal action changes from war to the operation and then to settlement as c_1 rises. Pairwise indifference gives the cutoffs in (3): $u_W = u_L$ at κ_W, $u_L = u_S$ at κ_L, and $u_W = u_S$ at κ_S.

The operation is optimal for some types if and only if $\kappa_W < \kappa_L$. If $\kappa_W \geq \kappa_L$, its payoff never exceeds $\max\{u_W, u_S\}$. The coercer then fights below κ_S and settles above it. □

B.4. Proof of Lemma 4

Proof. After concession, war occurs with probability $W(x)$ and gives the target $1 - p - c_2$. The operation occurs with probability $L(x) - W(x)$ and gives the target $1 - x - S(x)$. Settlement occurs otherwise and gives the target $1 - x$. Hence

$$V_2(x) = W(x)(1-h) + (1-x)(1-W(x)) - S(x)(L(x) - W(x))$$

The target's payoff after resistance is $V_2(0)$, evaluated at $x = 0$ and $S(0) = z$. The target concedes if and only if $V_2(x) \geq V_2(0)$. Rearranging yields (4). □

Remark 4 (Distributional Scope): Let $r(c) = \frac{f(c)}{1-F(c)}$ be the upper-tail hazard of a differentiable war-cost distribution. On the uncapped domain, the sufficient condition

$$\sup_{0 \leq c \leq \frac{p}{1-\lambda}} r(c) < \frac{1-\lambda}{h}$$

makes the target's acceptance margin strictly decreasing, so its accepted demands form an interval. For the uniform distribution this condition is exactly Assumption 1(ii). The continuation cutoffs, the target rule, Proposition 3, Proposition 7, and the threat–assurance identity do not require uniformity. The quadratic demands and uniqueness claims that use this interval argument do. Outside the hazard condition, the paper makes no connected-acceptance-set claim without a global solver.

Proof. Let $G(x) = V_2(x) - V_2(0)$ be the target's acceptance margin. In a region where the operation is dominated after compliance,

$$G'(x) = f(p-x)(h-x) - [1 - F(p-x)].$$

In an active, uncapped region,

$$G'(x) = f\Big(\kappa_{W(x)}\Big)\frac{h-x-z}{1-\lambda} - \Big[1 - F\Big(\kappa_{W(x)}\Big)\Big].$$

If either parenthesized harm term is nonpositive, the corresponding derivative is negative immediately. Otherwise divide by the positive survival probability. Since every relevant cutoff lies below $\frac{p}{1-\lambda}$ and each harm term is at most h, the stated upper-tail-hazard bound makes both derivatives strictly negative. The payoff, and hence G, is continuous where the operation enters or leaves the coercer's envelope, so the accepted set is an interval on the uncapped domain. For $F(c) = \frac{c}{\overline{c}}$, $r(c) = \frac{1}{\overline{c}-c}$; its supremum on the stated interval is $\frac{1}{\left[\overline{c}-\frac{p}{1-\lambda}\right]}$. Substitution reduces the bound to $\overline{c} > \frac{p+h}{1-\lambda}$. □

C. Proof of the Equilibrium Characterization

C.1. Proof of Proposition 1 and Proposition 2

Proof. Benchmark. Let $G^{\dagger}(x)$ be the target's payoff from concession minus its payoff from resistance. With interior cutoffs, the target concedes exactly when

$$x^2 + x(\overline{c} - p - h) - ha \leq 0$$

The constant term is negative, so the quadratic has one positive and one negative root. The target accepts every nonnegative demand up to the positive root and rejects every larger demand. The positive root is (1).

At this demand, war occurs with probability $F(p+a)$ after resistance and $F\Big(\kappa^{\dagger(x^{\dagger})}\Big)$ after concession. Definition 1 and the binding acceptance condition then give (2) and $\alpha^{\dagger} = 1 - F\Big(\kappa^{\dagger(x^{\dagger})}\Big)$.

Operation game. Let $G(x) = V_2(x) - V_2(0)$. First consider demands for which the operation is dominated after concession. Then $W(x) = L(x) = F(p-x)$, and the uniform distribution gives

$$G'(x) = \frac{h + p - \overline{c} - 2x}{\overline{c}} < 0$$

Assumption 1(ii) implies $\bar{c} > h + p$, so this derivative is negative. Next consider demands for which the operation is active after concession and $S(x) = z$. Then $L(x) = F\big(\frac{z}{\lambda}\big)$, and

$$G'(x) = \frac{f(\kappa_W)(h - x - z)}{1 - \lambda} - (1 - F(\kappa_W))$$

This derivative is negative if $\frac{h+p-2x-2z}{1-\lambda} < \bar{c}$. Its left side is no larger than $\frac{h+p}{1-\lambda}$, which is below $\bar{c}$ under the maintained restriction. The target's payoff is continuous where the operation enters the coercer's optimal menu. Thus G is strictly decreasing across the dominated and active regions wherever $S(x) = z$.

As $x \to 0^+$, the audience cost ceases to apply because the target has conceded. The operation–settlement cutoff therefore falls from $\kappa_L(0) = \frac{z+a}{\lambda}$ to $\frac{z}{\lambda}$. Although κ_W does not contain the audience cost, the probability of war can also jump if the operation is dominated on one side of $x = 0$, since κ_S contains a. Combining both changes gives

$$G(0^+) = z[L(0) - L(0^+)] + (h - z)[W(0) - W(0^+)]$$

Both bracketed differences are nonnegative. Removing the audience cost raises the settlement payoff and leaves the war and operation payoffs unchanged, so each type's optimal action after concession of 0^+ is weakly less forceful than after resistance. Hence $L(0) \geq L(0^+)$ and $W(0) \geq W(0^+)$. The coefficient $h - z$ multiplies a nonzero difference only when the operation is dominated at 0^+: if the operation is active on both sides of $x = 0$, then $\kappa_W(0) = \kappa_W(0^+) = \frac{p-z}{1-\lambda}$ and the second bracket vanishes. Domination at 0^+ requires $\kappa_W(0^+) \geq \kappa_L(0^+) = \frac{z}{\lambda}$, which rearranges to $z \leq \lambda p < h$, so the coefficient is positive whenever the difference is. Therefore $G(0^+) \geq z[L(0) - L(0^+)]$, with equality when the operation is active on both sides of $x = 0$. If it is active after both responses and $\kappa_L(0) \geq \bar{c}$, then $L(0) = 1$ and $L(0^+) = F\big(\frac{z}{\lambda}\big)$. Hence $G(0^+) = z\alpha$. Because $z > 0$, a positive demand is accepted if and only if $\alpha > 0$. The audience cost by itself does not determine this result.

Proposition 2 assumes $\alpha > 0$ and $S(x) = z$ at every accepted demand. Demands at which the bound binds are therefore rejected, and strict monotonicity over the remaining interval gives a unique largest accepted demand x^*. This also proves Remark 3. If $z \geq \lambda\bar{c}$, then $\alpha = 0$ and the quadratic's constant term $-(1 - \lambda)\bar{c}z\alpha$ is zero. Its roots are 0 and $-((1 - \lambda)\bar{c} - p +$

$2z - h$). No positive demand is then accepted, so $x^* = 0$ and $\Omega^* = 0$; conversely $\alpha > 0$ gives $G(0^+) = z\alpha > 0$ and a positive accepted demand.

Equilibrium strategies. Lemma 2 establishes that no coercer type prefers resistance. In the operation game, every type demands x^*, the target concedes, and types below $\kappa_W(x^*)$ fight, so war occurs with probability $W(x^*)$. In the benchmark, every type demands $x^\dagger$, and types below $\kappa^\dagger(x^\dagger)$ fight.

Intuitive criterion. Let $U_C(x, c_1)$ be a type's optimized continuation payoff after concession of x, and let $U_R(c_1)$ be its optimized payoff after resistance. In the benchmark these functions are

$$U_C^\dagger(x, c_1) = \max\{x, p - c_1\}, \quad U_R^\dagger(c_1) = \max\{-a, p - c_1\} \tag{11}$$

In the operation game they are

$$U_C(x, c_1) = \max\{x, p - c_1, \min\{x + z, 1\} - \lambda c_1\}, \quad U_R(c_1) = \max\{-a, p - c_1, z - \lambda c_1\} \tag{12}$$

In either game, $U_C(x, c_1)$ is weakly increasing in x. Lemma 2 gives $U_R(c_1) \leq U_C(x^e, c_1)$, where x^e denotes the relevant pooling demand.

Apply the continuous-type version of the intuitive criterion. For an off-path demand x', first remove a type if its equilibrium payoff exceeds the most it can receive under any target response that is optimal for some posterior. The target then places weight only on the remaining types. A candidate fails the criterion only if some remaining type strictly prefers x' under every best response to such beliefs.

First take $x' < x^e$. Monotonicity gives $U_C(x', c_1) \leq U_C(x^e, c_1)$, while Lemma 2 gives $U_R(c_1) \leq U_C(x^e, c_1)$. No target response can therefore make any type strictly better off. The positive-measure interval of types that fight after concession of x^e is not equilibrium dominated: these types also fight after either response to x'. A posterior supported on that interval makes the target indifferent, so the maintained tie rule selects concession, but no type profits.

Now take $x' > x^e$. Suppose first that concession is optimal for the target under some posterior. Because $U_C(x', c_1) \geq U_C(x^e, c_1)$ for every type, no type is equilibrium dominated. The prior is therefore an admissible off-path belief. Under the prior, the target strictly rejects x': x^e is the largest accepted demand, and the target concedes at equality. Rejection gives every type $U_R(c_1) \le U_C(x^e, c_1)$. If concession is not optimal under any posterior, rejection is optimal under every posterior; a belief supported on the equilibrium fighting types is admissible and again deters the deviation. Thus every off-path demand has an intuitive-criterion-consistent belief and sequentially rational response under which no type gains. Both pooling equilibria satisfy the criterion.

D1. The stronger D1 test also compares types that survive equilibrium dominance. Let $q \in [0, 1]$ be the probability that the target concedes after x'. For each type, let $\mathcal{D}_{c_1}$ contain the values of q that make the deviation strictly profitable and let $\mathcal{D}^0_{c_1}$ contain those that make it payoff-equivalent. D1 prunes c_1 if $\mathcal{D}_{c_1} \cup \mathcal{D}^0_{c_1}$ is a proper subset of $\mathcal{D}_{\hat{c}_1}$ for some other type $\hat{c}_1$.

In the benchmark, take $x' = x^\dagger + \varepsilon < p$ and define $k = p - x^\dagger$ and $k' = p - x'$. Types $c_1 \le k'$ fight after either target response, so $\mathcal{D}_{c_1} = \emptyset$ and $\mathcal{D}^0_{c_1} = [0, 1]$; D1 cannot prune them. Types in $(k', k]$ fight on the equilibrium path and after resistance but settle after concession of x', so $\mathcal{D}_{c_1} = (0, 1]$ and $\mathcal{D}^0_{c_1} = \{0\}$. Types above k require a strictly positive concession probability and are pruned by the switching types. The D1-admissible support is therefore $[0, k]$, not only the switching interval. The target is indifferent for the always-fighting types and strictly prefers concession for the switching types because $x' < p \le p + c_2$. Under the maintained tie rule, every D1-admissible belief induces concession, and the switching types profit.

The same calculation applies in the operation game. Choose $x' = x^* + \varepsilon$ so that $x' + z < p$, and define $k = \kappa_W(x^*)$ and $k' = \kappa_W(x')$. Types below k' fight after either response and cannot be pruned. Types in $(k', k]$ fight on the equilibrium path and after resistance but use the operation after concession of x'. Types above k require a strictly positive concession probability and are pruned by the switching types. D1 therefore permits beliefs on $[0, k]$. Concession leaves the target indifferent for the always-fighting types and strictly improves its payoff for the switching types because $x' + z < p \le p + c_2$. The tie rule again induces concession under every admissible belief, so the switching types profit.

If $c_2 > 0$, neither conclusion depends on the tie rule. In the benchmark choose $x' \in (p, p + c_2)$; in the operation game choose $x' \in (p - z, p + c_2 - z)$. Every type in the corresponding D1 support then switches away from war after concession, and the target strictly prefers concession. If $c_2 = 0$ and rejection is instead allowed at off-path indifference, a posterior on types that fight after either response can support rejection. Thus the pooling equilibria fail D1 under the maintained convention, but D1′s conclusion at $c_2 = 0$ is tie-breaking-sensitive. The paper imposes the intuitive criterion rather than D1. Proposition 8 extends this calculation to every equilibrium with pure target responses.

Closed form in the operation game. In the active, interior region, $W = \frac{p-x-z}{(1-\lambda)\overline{c}}$, $W_0 = \frac{p-z}{(1-\lambda)\overline{c}}$, and $W_0 - W = \frac{x}{(1-\lambda)\overline{c}}$. Moreover, $L = \frac{z}{\lambda\overline{c}} = 1 - \alpha$ is constant in x. If $\kappa_L(0) \geq \overline{c}$, every nonfighting type uses the operation after resistance. Hence $L_0 = 1$ and $L_0 - L = \alpha$. The binding acceptance condition becomes

$$\frac{x((1-\lambda)\overline{c} - p + x + z)}{(1-\lambda)\overline{c}} = \frac{x(h-z)}{(1-\lambda)\overline{c}} + z\alpha$$

Rearranging yields $x^2 + x((1-\lambda)\overline{c} - p + 2z - h) - (1-\lambda)\overline{c}z\alpha = 0$. Its constant term is negative because $\alpha > 0$. The unique positive root is therefore (5). $\square$

D. Proofs of the Comparative Statics

D.1. Proof of Remark 1

Proof. At the largest accepted demand, (4) binds, so $V_2(x^*) = V_2(0)$. Let $W = W(x^*)$, $L = L(x^*)$, $W_0 = W(0)$, and $L_0 = L(0)$. Substituting the target's payoffs yields

$$W(1-h) + (1-x^*)(1-W) - S(x^*)(L-W) = W_0(1-h) + (1-W_0) - z(L_0 - W_0)$$

Cancel common terms and collect the terms in x^*:

$$x^*(1-W) = h(W_0 - W) + z(L_0 - W_0) - S(x^*)(L-W)$$

Separating hW_0 from hW gives (6). Definition 1 gives $\Omega^* = x^*(1-W)$. Therefore coercive ability is zero if and only if $\Theta^* = \Delta^*$. $\square$

D.2. Proof of Proposition 3

Proof. Suppose the operation is active after concession and the operation–settlement cutoff is interior. By Lemma 3, the coercer settles when $c_1 > \kappa_L(x^*)$. Thus $\alpha = 1 - F(\kappa_L(x^*))$.

With a general benefit $S(y)$ and additional cost $C(y)$, settlement pays y, while the operation pays $y + S(y) - \lambda c_1 - C(y)$. The transfer cancels from their difference. The actions are equally valuable at $\kappa_L(y) = \frac{S(y)-C(y)}{\lambda}$, with derivative $\frac{\partial \kappa_L}{\partial y} = \frac{S'(y)-C'(y)}{\lambda}$. In the benchmark, the corresponding war–settlement cutoff is $\kappa^\dagger(y) = p - y$, whose derivative is -1. Each additional unit transferred therefore lowers the cutoff one-for-one and makes settlement more likely.

Under the present specification, Assumption 1(i) makes $S(y) = z$ at every accepted demand, and there is no additional cost, so $C(y) = 0$. Consequently, $\frac{\partial \kappa_L}{\partial y} = 0$, $\kappa_L(x^*) = \frac{z}{\lambda}$, and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$. Since F is strictly increasing at this cutoff, $\frac{\partial \alpha}{\partial z} = -\frac{f\left(\frac{z}{\lambda}\right)}{\lambda} < 0$ and $\frac{\partial \alpha}{\partial \lambda} = f\left(\frac{z}{\lambda}\right)\frac{z}{\lambda^2} > 0$.

The zero derivative is specific to the fixed-effect specification. If $S(y) = z(1-y)$ or operating cost rises with the transfer, then $\frac{\partial \kappa_L}{\partial y} < 0$. A larger transfer would make settlement more likely, although less rapidly than in the benchmark. $\square$

D.3. Proof of Corollary 1

Proof. Settlement after transfer y pays $y - a\mathbb{1}[y=0]$ and the operation pays $y + g - \lambda c_1$. Their difference is $g + a\mathbb{1}[y=0] - \lambda c_1$, which vanishes at $\kappa_L(y) = \frac{g+a\mathbb{1}[y=0]}{\lambda}$. The target's loss d does not enter the coercer's comparison and therefore does not appear in the cutoff. After compliance $\kappa_L(x) = \frac{g}{\lambda}$ and $\alpha = 1 - F\left(\frac{g}{\lambda}\right)$, which is zero exactly when $g \geq \lambda\bar{c}$.

Now set $g = 0$. With no share conveyed, the operation pays the coercer $y - \lambda c_1$ after transfer y against y from settlement, so every type with $c_1 > 0$ settles after compliance: $\alpha = 1$, $W(x) = L(x) = F(p - x)$, and the corner of Remark 3, which requires the operation–settlement cutoff to reach the top of the type support, is unreachable. After resistance the operation pays $-\lambda c_1$ against $-a$ from settlement and $p - c_1$ from war, so $\kappa_W(0) = \frac{p}{1-\lambda}$ and $\kappa_L(0) = \frac{a}{\lambda}$, and the operation is active if and only if $a > \lambda(p + a)$. Because the games share the post-compliance node, the target's loss from conceding x is the same strictly increasing function in both, so the largest accepted demand rises with expected loss after resistance. With interior post-resistance cutoffs that loss is $\frac{h(p+a)}{\bar{c}}$ in the benchmark and $\frac{hp}{(1-\lambda)\bar{c}} + \frac{d\left(\frac{a}{\lambda} - \frac{p}{1-\lambda}\right)}{\bar{c}}$ with the operation, and their difference has the sign of $(a(1-\lambda) - \lambda p)(d - \lambda h)$, whose first factor is positive exactly when the operation is active after resistance. The capability therefore raises the accepted demand and coercive ability if and only if $d > \lambda h$. $\square$

D.4. Proof of Proposition 6

Proof. Benchmark. Types below $\kappa^\dagger = p - x^\dagger$ fight and receive $p - c_1$. The remaining types settle and receive $x^\dagger$. Under the uniform distribution,

$$V^\dagger = \frac{1}{\bar{c}}\left(\int_0^{\kappa^\dagger} (p-c)\,\mathrm{d}c + \int_{\kappa^\dagger}^{\bar{c}} x^\dagger\,\mathrm{d}c\right) = x^\dagger + \frac{1}{\bar{c}}\kappa^\dagger\left(p - \frac{\kappa^\dagger}{2} - x^\dagger\right)$$

Since $p - x^\dagger = \kappa^\dagger$, the term in parentheses is $\frac{\kappa^\dagger}{2}$. Substitution gives the first line of (10).

Operation game. Types below κ_W fight. Types in $(\kappa_W, \kappa_L]$ use the operation and receive $x^* + z - \lambda c_1$. The remaining types settle. Integrating over the three regions gives

$$V^* = x^* + \frac{1}{\bar{c}}\left(\kappa_W(p - x^* - z) - (1-\lambda)\frac{\kappa_W^2}{2} + z\kappa_L - \lambda\frac{\kappa_L^2}{2}\right)$$

Since $p - x^* - z = (1-\lambda)\kappa_W$, the first two terms reduce to $\frac{(1-\lambda)\kappa_W^2}{2}$. Since $\kappa_L = \frac{z}{\lambda}$, the remaining operation term becomes $\frac{z^2}{2\lambda}$. Substituting $\kappa_W = \frac{p-x^*-z}{1-\lambda}$ yields the second line of (10).

Define Q_0 and Q_1 as in the main text. An interior operation-game war cutoff at x^* implies $p - x^* > z > 0$, while Assumption 1(ii) implies $p - x^* < \bar{c}$. The benchmark cutoff evaluated at x^* is therefore interior. Adding and subtracting $Q_0(x^*)$ gives the main-text decomposition. Simplification gives

$$Q_1(x^*) - Q_0(x^*) = \frac{(\lambda(p-x^*) - z)^2}{2\lambda(1-\lambda)\bar{c}} \geq 0$$

which establishes the nonnegative menu gain. Moreover, $Q_{0'}(y) = 1 - \frac{p-y}{\bar{c}} > 0$ whenever the benchmark cutoff is interior. Thus $Q_0(x^*) - Q_0(x^\dagger) < 0$ if $x^* < x^\dagger$. The equilibrium payoff falls exactly when this loss exceeds the menu gain. □

Proposition 7 (The Post-Compliance Cutoff Is Selection-Free): Consider any equilibrium and any demand conceded with positive probability. If the operation is active after that demand and $S(x) = z$, the coercer settles if and only if $c_1 > \frac{z}{\lambda}$. This cutoff is the same at every such demand and under every off-path belief. If the operation is active after every conceded demand, so every accepted demand exceeds $p - \frac{z}{\lambda}$, then the settling types are $\left(\frac{z}{\lambda}, \bar{c}\right]$ in every equilibrium.

D.5. Proof of Proposition 7

Proof. Fix an equilibrium and a transfer $y = x > 0$ conceded with positive probability. The coercer then chooses among settlement, war, and the operation, with payoffs x, $p - c_1$, and $x + S(x) - \lambda c_1$. These payoffs depend only on c_1, x, and the model parameters. Because the target has already moved, they do not depend on its beliefs, its mixing probability, or the demands made by other types.

The operation payoff minus the settlement payoff is $S(x) - \lambda c_1$. When $S(x) = z$, this difference is $z - \lambda c_1$ and does not depend on x. The operation is preferred to settlement exactly when $c_1 < \frac{z}{\lambda}$. Because the operation is active by hypothesis, Lemma 3 implies that types above $\kappa_L(x) = \frac{z}{\lambda}$ settle and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$.

The cutoff is therefore $\frac{z}{\lambda}$ at every conceded demand for which the operation is active. Activity itself depends on the demand. Equation (3) implies that the operation enters the coercer's envelope after concession of x exactly when $x > p - \frac{z}{\lambda}$. In a separating equilibrium with two accepted demands, the operation may be active after the larger demand and dominated after the smaller one. The corresponding settlement cutoffs would then be $\frac{z}{\lambda}$ and $\kappa_S(x_L)$. The common settlement probability $1 - F\left(\frac{z}{\lambda}\right)$ applies when every accepted demand exceeds $p - \frac{z}{\lambda}$. □

D.6. Proof of The Resistance-Stage War Comparison

Proof. If the operation is active after resistance, the coercer fights when $c_1 \le \kappa_W(0)$. If it is dominated, the coercer fights when $c_1 \le \kappa_S(0) = p + a$. After resistance, $S(0) = z$ and the audience cost applies. Thus $\kappa_W(0) = \frac{p-z}{1-\lambda}$ and $\kappa_L(0) = \frac{z+a}{\lambda}$, which gives the two probabilities stated in the main text. The operation is active if and only if $\kappa_W(0) < \kappa_L(0)$, or

$$\lambda(p - z) < (1 - \lambda)(z + a)$$

Rearranging gives $\lambda(p + a) < z + a$. The condition for the operation to lower war is $\kappa_W(0) < \kappa_S(0) = p + a$, which rearranges to the same inequality. Thus the operation is active exactly when it lowers war. Assumption 1(ii) implies $\kappa_W(0) \le \frac{p}{1-\lambda} < \bar{c}$, so the reduction is strict whenever the operation enters the menu. The operation never raises war after resistance.

The extension in which the operation carries its own audience cost is derived in §F.3. The Operation's Audience Cost. $\square$

D.7. Proof of Proposition 4

Proof. Benchmark. With an interior war cutoff, $W^\dagger\big(x^\dagger\big) = \frac{p-x^\dagger}{\overline{c}}$. Therefore

$$\Omega^\dagger = x^\dagger\big(1-W^\dagger\big) = \frac{x^\dagger\big(\overline{c}-p+x^\dagger\big)}{\overline{c}}$$

Equation (1) implies $x^{\dagger 2} = ha - x^\dagger(\overline{c}-p-h)$. Substitute this expression into the numerator:

$$x^\dagger(\overline{c}-p) + x^{\dagger 2} = x^\dagger(\overline{c}-p) + ha - x^\dagger(\overline{c}-p-h) = h\big(x^\dagger + a\big)$$

Dividing by $\overline{c}$ gives $\Omega^\dagger = \frac{h(x^\dagger+a)}{\overline{c}}$.

Operation game. In the active, interior region, $W(x^*) = \frac{p-x^*-z}{(1-\lambda)\overline{c}}$. Hence $\Omega^* = \frac{x^*((1-\lambda)\overline{c}-p+x^*+z)}{(1-\lambda)\overline{c}}$. Equation (5) gives $x^{*2} = (1-\lambda)\overline{c}z\alpha - x^*((1-\lambda)\overline{c}-p+2z-h)$. Substituting yields

$$\begin{aligned} &x^*((1-\lambda)\overline{c}-p+z) + x^{*2} \\ &\quad = x^*((1-\lambda)\overline{c}-p+z) + (1-\lambda)\overline{c}z\alpha \\ &\qquad - x^*((1-\lambda)\overline{c}-p+2z-h) \\ &\quad = x^*(h-z) + (1-\lambda)\overline{c}z\alpha \end{aligned}$$

Divide by $(1-\lambda)\overline{c}$ to obtain $\Omega^* = \frac{x^*(h-z)}{(1-\lambda)\overline{c}} + z\alpha$. Comparing this expression with the benchmark gives (8). $\square$

D.8. Proof of Proposition 5

Proof. Define $\varphi(t) = t^2 + [(1-\lambda)\overline{c}-p+2z-h]t - (1-\lambda)\overline{c}z\alpha$. Its unique positive root is x^*. Assumption 1(ii) implies $(1-\lambda)\overline{c} > p+h$, so $\varphi(-z) = z(p+h-(1-\lambda)\overline{c}-z-(1-\lambda)\overline{c}\alpha) < 0$. Since $x^\dagger - z \geq -z$, this argument cannot lie below the negative root. Therefore $x^\dagger > x^* + z$ if and only if $\varphi\big(x^\dagger - z\big) > 0$.

Expand $\varphi\big(x^\dagger - z\big)$ and use $x^{\dagger 2} = ha - (\overline{c}-p-h)x^\dagger$ from (1):

$$
\begin{aligned}
\varphi\big(x^\dagger - z\big) = ha + z^2 - [(1-\lambda)\bar{c} - p + 2z - h]z \\
-(1-\lambda)\bar{c}z\alpha \\
+x^\dagger[(1-\lambda)\bar{c} - p + 2z - h \\
-(\bar{c} - p - h) - 2z]
\end{aligned}
$$

The coefficient on $x^\dagger$ simplifies to $-\lambda\bar{c}$. Collecting the remaining terms gives

$$
\varphi\big(x^\dagger - z\big) = ha + z(p + h - (1-\lambda)\bar{c} - z - (1-\lambda)\bar{c}\alpha) - \lambda\bar{c}\,x^\dagger
$$

Positivity of this expression, written in primitives, is

$$
(p + c_2)a + z(2p + c_2 - (1-\lambda)\bar{c} - z - (1-\lambda)\bar{c}\alpha) > \lambda\bar{c}\,x^\dagger
$$

Dividing $\varphi\big(x^\dagger - z\big) > 0$ by $(1-\lambda)\bar{c}$ and substituting $1 - W\big(x^\dagger - z\big) = \frac{(1-\lambda)\bar{c} - p + x^\dagger}{(1-\lambda)\bar{c}}$ gives (9).

Finally, compare each coercer type's payoff. The same condition gives $x^\dagger > x^* + z$, so for every $c_1 \geq 0$,

$$
\max\{x^*, p - c_1, x^* + z - \lambda c_1\} \leq \max\big\{x^\dagger, p - c_1\big\}
$$

The inequality is strict when $p - c_1 < x^\dagger$. These are precisely the types that settle after compliance in the benchmark. They have positive measure because the benchmark cutoff is interior.

War after compliance. By (3) and Lemma 1, $\kappa_W(x^*) > \kappa^\dagger\big(x^\dagger\big)$ is equivalent to $\frac{p - x^* - z}{1-\lambda} > p - x^\dagger$ and hence to $\big(x^\dagger - x^* - z\big) + \lambda\big(p - x^\dagger\big) > 0$. The first addend is strictly positive because $x^\dagger > x^* + z$ and the second is strictly positive because the benchmark post-compliance war cutoff is interior. Both cutoffs lie strictly inside $[0, \bar{c}]$ under the stated hypotheses, so F is strictly increasing between them and $W(x^*) = F(\kappa_W(x^*)) > F\big(\kappa^\dagger\big(x^\dagger\big)\big)$. □

D.9. Interior Switch Accounting

Hold transfer y fixed, suppose $S(y) = z$, and let every continuation cutoff be interior. Write $A(y) = a\mathbb{1}[y = 0]$ and define the operation's entry margin

$$
M(y) = z + (1-\lambda)A(y) - \lambda(p - y).
$$

The operation enters the coercer's menu when $M(y) > 0$. Relative to the benchmark, the masses that switch from war and settlement to the operation are

$$m_{W(y)} = \frac{M(y)}{(1-\lambda)\bar{c}}, \quad m_{S(y)} = \frac{M(y)}{\lambda\bar{c}}.$$

The first equality follows from $\kappa_{S(y)} - \kappa_{W(y)} = \frac{M(y)}{1-\lambda}$ and the second from $\kappa_{L(y)} - \kappa_{S(y)} = \frac{M(y)}{\lambda}$.

A former fighter lowers the target's loss by $h - z$, while a former settler raises it by z. The change in expected harm at history y is therefore

$$zm_{S(y)} - (h-z)m_{W(y)} = \frac{M(y)(z-\lambda h)}{\lambda(1-\lambda)\bar{c}}. \tag{13}$$

This expression has the sign of $z - \lambda h$. Applying it after both target responses and using $M(0) - M(x) = (1-\lambda)a - \lambda x$ shows that the change in threat relative to the change in assurance has the sign of $(z - \lambda h)((1-\lambda)a - \lambda x)$. The first factor determines whether the operation raises harm at a given history, and the second determines whether more types switch after resistance or compliance.

The closed-form region truncates the settlement-to-operation switch after resistance because $\kappa_{L(0)} \geq \bar{c}$. There, $\Theta^* < \Theta^\dagger$ and $\Delta^* > \Delta^\dagger$ hold together exactly when

$$(1-\lambda)(\bar{c}-p-a)z < M(0)(h-z), \quad \frac{M(x^*)(z-\lambda h)}{\lambda(1-\lambda)} + h\big(x^\dagger - x^*\big) > 0. \tag{14}$$

The first inequality says that war replacement reduces punishment after resistance despite the operation's effect on former settlers. The second says that post-compliance harm rises once the smaller accepted demand is included. This is an exact conditional statement, not a claim about prevalence in a parameter population.

E. Separating Demands and D1

Lemma 5 (Separating Demands Do Not Change the Selected Outcome): Consider either game under the conditions of Proposition 1 or Proposition 2 and any perfect Bayesian equilibrium in which the target's response is a pure function of the demand. Let $\overline{x}$ be the largest demand the target accepts and let x^e denote $x^\dagger$ in the benchmark and x^* in the operation game. Every type that does not fight after concession of $\overline{x}$ demands $\overline{x}$. Only types that fight after both target responses may choose another demand, and their separation does not change the target's acceptance condition at $\overline{x}$. Hence $\overline{x} \leq x^e$. The pooling equilibrium attains x^e and is therefore coercer-optimal even when separating demands are allowed.

Proof. Conceding zero weakly improves the target's payoff, so the accepted set is nonempty. Lemma 2 and Assumption 1(i) imply that each type's payoff after concession is weakly increasing in the demand and strictly increasing for any type that does not fight afterward. Assumption 1(ii) leaves a positive measure of such types at every accepted demand. If the accepted set had no maximum, these types would have no optimal demand. It therefore has a largest element $\overline{x}$, and every type that does not fight after its concession must demand it.

A type that fights after concession of $\overline{x}$ also fights after resistance by Lemma 2. Such a type gives the target payoff $1 - h$ after either response, so it contributes zero to the target's payoff difference between concession and resistance. All other types demand $\overline{x}$. Bayes' rule therefore implies that the sign of the target's payoff difference at $\overline{x}$ is the same under its on-path posterior as under the prior: separating always-fighting types changes only terms whose difference is zero. Thus any largest demand accepted in a separating equilibrium is also accepted under the prior and cannot exceed the largest prior-accepted demand x^e. Propositions 1 and 2 show that the pooling equilibrium attains x^e. Because every type's continuation payoff is weakly increasing in an accepted demand, no separating equilibrium makes any type better off. $\square$

The D1 calculation in the proof of Proposition 1 and Proposition 2 is local to the two pooling equilibria. The same comparison applies to every equilibrium in the class the paper compares, and its verdict is the same in both games.

PROPOSITION 8 (No Accepted Demand Survives D1 Under Pure Target Responses): Consider either game under Assumption 1 and the rule that the target concedes when indifferent. Take any perfect Bayesian equilibrium in which the target's response is a pure function of the demand, and let $\overline{x}$ be the largest demand the target accepts. Every coercer type's equilibrium payoff is then $U_C(\overline{x}, c_1)$, so the equilibrium with the largest accepted demand is the one every type weakly prefers within this class. If the war cutoff after concession of $\overline{x}$ is interior, the equilibrium fails D1. If $c_2 > 0$, the failure does not depend on the tie rule.

E.1. Proof of Proposition 8

Proof. Write $\kappa(x)$ for the war cutoff after concession of x: it equals $\kappa_W(x)$ where the operation is active after that demand, $\kappa_S(x) = p - x$ where the operation is dominated, and $\kappa^\dagger(x) = p - x$ in the benchmark. On an uncapped local interval the branch expressions agree where the operation enters or leaves the envelope, so κ is continuous and strictly decreasing there. Lemma 2 gives $\kappa(x) \leq \kappa(0)$, so a type that fights after concession also fights after resistance. Assumption 1(ii) gives $\kappa(x) \leq \frac{p}{1-\lambda} < \overline{c}$, so at every demand a positive measure of types does not fight after concession.

The largest accepted demand and the equilibrium payoff. Conceding 0 raises each type's settlement payoff from $-a$ to 0 and leaves its war and operation payoffs unchanged, so each type's optimal action after concession of 0 is weakly less forceful than after resistance and the target's payoff from conceding 0 is weakly higher than from resisting. Under the tie rule the target concedes 0, so the accepted set is nonempty. Lemma 2 and the monotonicity of $U_C(x, c_1)$ in x imply that every type weakly prefers any accepted demand to any rejected demand, strictly so for types that do not fight after concession. Those types have positive measure, and their payoff is strictly increasing in the accepted demand and strictly above their payoff from any rejected demand, so they would have no optimal demand if the accepted set had no maximum. The accepted set therefore attains a largest element $\overline{x}$, which they demand. Optimality then gives $u^*(c_1) = U_C(\overline{x}, c_1)$ for every type: for types pooled at $\overline{x}$ by construction; for a type at a smaller accepted demand because monotonicity and optimality force equality; and for a type at a rejected demand because optimality gives $U_R(c_1) \geq U_C(\overline{x}, c_1)$ while Lemma 2 gives the reverse.

Pruning at an upward deviation. Let $k = \kappa(\overline{x})$ and suppose $0 < k < \overline{c}$. An interior k implies $\overline{x} + z < p$: if the operation is active after $\overline{x}$, then $k > 0$ is $p - \overline{x} - z > 0$; if it is dominated, then $\kappa_W(\overline{x}) \geq \kappa_L(\overline{x}) = \frac{z}{\lambda}$ gives $z \leq \lambda(p - \overline{x}) < p - \overline{x}$. Choose $\varepsilon > 0$ small enough that $x' = \overline{x} + \varepsilon$ satisfies $x' + z < p$ and $k' = \kappa(x') > 0$. The argument establishing $S(x) = z$ above then makes κ continuous and strictly decreasing from $\overline{x}$ to x', so $k' < k$. Let $q \in [0, 1]$ be the probability that the target concedes after x'. Type c_1's payoff from the deviation is $qU_C(x', c_1) + (1 - q)U_R(c_1)$, so the deviation is strictly profitable exactly when $q\Delta(c_1) > E(c_1)$, where

$$\Delta(c_1) = U_C(x', c_1) - U_R(c_1), \quad E(c_1) = U_C(\overline{x}, c_1) - U_R(c_1)$$

Monotonicity gives $\Delta(c_1) \geq E(c_1) \geq 0$, and Lemma 2 gives $E(c_1) = 0$ exactly for $c_1 \leq k$ and $\Delta(c_1) = 0$ exactly for $c_1 \leq k'$. Three groups follow. Types $c_1 \leq k'$ fight after either response to x', so $\mathcal{D}_{c_1} = \emptyset$ and $\mathcal{D}^0_{c_1} = [0, 1]$. Types in $(k', k]$ have $\mathcal{D}_{c_1} = (0, 1]$ and $\mathcal{D}^0_{c_1} = \{0\}$. Types above k have $E(c_1) > 0$, so $\mathcal{D}_{c_1} \cup \mathcal{D}^0_{c_1} = \left[\frac{E(c_1)}{\Delta(c_1)}, 1\right]$ with $\frac{E(c_1)}{\Delta(c_1)} > 0$, a proper subset of $(0, 1]$. The first two groups have $\mathcal{D}_{c_1} \cup \mathcal{D}^0_{c_1} = [0, 1]$, which no $\mathcal{D}_{\hat{c}_1}$ can properly contain, so D1 cannot prune them; the third group is pruned by the second, which has measure $\frac{k - k'}{\overline{c}} > 0$. Every D1-admissible belief is therefore supported on $[0, k]$.

The target's reply. Under any belief on $[0, k]$, types in $[0, k']$ fight after concession of x' and after resistance, contributing $1 - h$ to both. Types in $(k', k]$ fight after resistance but not after concession of x'; they take at most $x' + z < p \leq h$, so each raises the target's payoff from conceding above $1 - h$. Conceding is therefore weakly better than resisting, and strictly better when the belief places mass on $(k', k]$. Under the tie rule the target concedes.

Conclusion. If x' is off the equilibrium path, D1 confines the belief to $[0, k]$, the target concedes, and every type in $(k', k]$ receives $U_C(x', c_1) > U_R(c_1) = u^*(c_1)$, contradicting equilibrium. If x' is on the path, it is rejected because $\overline{x}$ is the largest accepted demand; every type that plays it must satisfy $U_R(c_1) = U_C(\overline{x}, c_1)$, hence $E(c_1) = 0$ and $c_1 \leq k$, so the belief obtained from Bayes' rule is also supported on $[0, k]$. The computation above then makes conceding weakly better, and the tie rule requires concession, contradicting rejection. Either way the equilibrium fails D1.

Independence of the tie rule. Suppose $c_2 > 0$ and take $x' \in (p - z, p + c_2 - z)$ in the operation game and $x' \in (p, p + c_2)$ in the benchmark. Since $\overline{x} + z < p$, such an x' exceeds $\overline{x}$. In the operation game $x' > p - z$ gives $\lambda(p - x') < \lambda z < z$, so the operation is active after concession of x' and $\kappa(x') = \kappa_W(x') < 0$; in the benchmark $x' > p$ gives $\kappa^{\dagger}(x') < 0$. No admissible type then fights after concession of x', every admissible type takes at most $x' + z < h$, and the target strictly prefers conceding under every belief supported on $[0, k]$, whether that belief comes from D1 or from Bayes' rule. The switching types again profit.

The excluded boundary. If $k \leq 0$, no type fights after concession of $\overline{x}$, the group $(k', k]$ is empty, and D1 retains only the types that minimize E. The argument does not apply, and a pure-response equilibrium can survive: if $p + a \geq \overline{c}$ in the benchmark, every type fights after resistance, the target accepts exactly the demands $x \leq h$, and at $\overline{x} = h$ the D1-admissible belief is concentrated on $c_1 = 0$, against which the target strictly refuses every larger demand. Propositions 1 and 2 exclude this boundary by assuming interior war cutoffs. $\square$

F. Extensions and Scope

F.1. Escalation and Recontestation

The following remark shows how escalation risk changes the operation–settlement boundary.

Remark 5 (Escalation Risk Partially Restores Assurance): Suppose the operation triggers general war with probability $q \in [0, 1)$, and that general war reopens the dispute as it does elsewhere in the model. The coercer pays λc_1 whenever it uses the operation and pays c_1 in addition when escalation occurs, so after transfer y the operation pays $(1-q)(y+S(y)) + qp - \lambda c_1 - qc_1$ against $y - a\mathbb{1}[y=0]$ from settlement. If the operation is active after compliance and $S(y) = z$, the operation–settlement cutoff is (15). Its derivative in the transfer is $\frac{\partial \kappa_L^q}{\partial y} = -\frac{q}{\lambda+q} < 0$. Its derivative in the escalation probability, evaluated at $q = 0$, is $\frac{\partial \kappa_L^q}{\partial q} = \frac{\lambda(p-y-z)-z}{\lambda^2}$, which is negative wherever the operation is active after compliance because activity requires $z > \lambda(p - y)$. Setting $q = 0$ recovers $\kappa_L(y) = \frac{z}{\lambda}$.

$$\kappa_L^q(y) = \frac{(1-q)z + q(p-y)}{\lambda + q} \tag{15}$$

The remark lets the operation trigger general war with probability q. The next section lets general war reopen only a share of an accepted concession. Either change alters the target's acceptance constraint. The first is signed only locally; the second is solved in closed form. We make no claim that the baseline ordering survives arbitrary escalation or recontestation technologies.

F.2. Appropriative War

The baseline lets general war reopen the whole dispute after either target response, so an accepted transfer is worth nothing to the coercer once war occurs. This section replaces that assumption with its opposite pole and every point between. Let general war after compliance reclaim only a share of the concession and contest the remainder of the good. Nothing has been handed over after resistance, so that node is unchanged and the extension operates on the post-compliance branch alone.

Remark 6 (Appropriative War): Let $\tau \in [p, 1]$ index how far an accepted transfer deters war after compliance: war then pays the coercer $p + (1-\tau)x - c_1$ and the target $1 - h - (1-\tau)x$. If war reclaims a share ρ of the transfer and contests the rest of the good, then $\tau = \rho + (1-\rho)p$. The baseline is $\tau = 1$; war reclaims nothing at $\tau = p$, where every post-compliance action leaves the transfer in place. After compliance, each war cutoff is the baseline's with τx in place of x,

$$\kappa^{\dagger}(x) = p - \tau x, \quad \kappa_W(x) = \frac{p - \tau x - z}{1 - \lambda}$$

while $\kappa_L(x) = \frac{z}{\lambda}$ and $\alpha = 1 - F\left(\frac{z}{\lambda}\right)$ are unchanged, as is the entire post-resistance node. The target's expected payment is $x(1 - \tau W(x))$ and still equals $\Theta^* - \Delta^*$ for Θ^* and Δ^* as defined in (6). The largest accepted demands are the positive roots of (16), and

$$\Omega^{\dagger} = \frac{h\left(\tau x^{\dagger} + a\right)}{\bar{c}}, \quad \Omega^* = \frac{\tau x^*(h - z)}{(1-\lambda)\bar{c}} + z\alpha$$

Setting $\tau = 1$ recovers (1), (5), and Proposition 4.

$$\tau^2 x^2 + (\bar{c} - \tau(p+h))x - ha = 0, \quad \tau^2 x^2 + ((1-\lambda)\bar{c} - \tau(p+h) + 2\tau z)x - (1-\lambda)\bar{c}z\alpha = 0 \tag{16}$$

The cutoffs follow from the payoffs. After compliance, war pays $p + (1-\tau)x - c_1$, settlement pays x, and the operation pays $x + z - \lambda c_1$, so war beats settlement when $c_1 < p - \tau x$ and beats the operation when $(1-\lambda)c_1 < p - \tau x - z$. Neither settlement nor the operation involves war, so their comparison is the baseline's and Proposition 3 applies unchanged: the transfer cancels, κ_L is constant in the demand, and compliance still buys no restraint on the operation margin. What the transfer buys is war avoidance, and τ is the rate at which it does so. The target's loss from complying is $x(1 - W(x)) + (h + (1-\tau)x)W(x) + z(L(x) - W(x))$, which equals $x(1 - \tau W(x)) + hW(x) + z(L(x) - W(x))$. Equating this to the unchanged loss from resisting and substituting $W(0) - W(x) = \tau x/((1-\lambda)\bar{c})$ gives the second quadratic; setting $z = 0$ and $W = L = F\left(\kappa^{\dagger}\right)$ gives the first. The Assumption 1(ii) is unchanged and remains sufficient, because it binds most tightly at $\tau = 1$.

Two comparisons change. The demand enters coercive ability only through $\tau x^\dagger$ and τx^*, so the demand response that drives the result is damped but not removed. War after compliance also responds less to the demand, because the post-compliance war cutoff now moves by τ rather than by one for each unit conceded.

Remark 7 (Payoffs and War Under Appropriative War): Fix $\tau \in [p, 1]$ and suppose the premises of Proposition 5 hold with the cutoffs of the preceding remark. Every coercer type is weakly worse off with the operation if and only if (17) holds, and war after compliance is strictly more likely if and only if (18) holds. When $\tau < 1$ the second condition is the stronger of the two. When $\tau < 1$ the coercer's war payoff after compliance rises in the transfer, so types that fight in both games lose $(1-\tau)\big(x^\dagger - x^*\big)$ and the payoff loss is strict for every type rather than for a positive measure. At $\tau = 1$ those types are indifferent and the two conditions coincide, which is Proposition 5.

$$x^\dagger + \lambda\kappa^\dagger\big(x^\dagger\big) \geq x^* + z \tag{17}$$

$$\tau\big(x^\dagger - x^*\big) + \lambda\kappa^\dagger\big(x^\dagger\big) > z \tag{18}$$

Both conditions come from the same comparison. A benchmark settler receives $x^\dagger$; in the operation game it receives at most $\max\{x^*, x^* + z - \lambda c_1\}$ once c_1 exceeds $\kappa^\dagger\big(x^\dagger\big)$, and the operation branch is largest at that cutoff, which gives (17). Below the cutoff the benchmark type fights, and the same comparison delivers the same inequality, so (17) is necessary as well as sufficient. Rearranging $\kappa_W(x^*) > \kappa^\dagger\big(x^\dagger\big)$ gives (18). The condition stated in Proposition 5, $x^\dagger > x^* + z$, implies both at $\tau = 1$ but neither is implied by it in reverse: the slack term $\lambda\kappa^\dagger\big(x^\dagger\big)$ is the concession a benchmark settler at the war margin already forgoes, and the proposition's condition sets it aside. Proposition 5 is therefore stated on a sufficient condition, not a necessary one, at every τ.

F.3. The Operation's Audience Cost

The main text states the two activity boundaries for an operation that carries its own audience cost. The following remark and derivation are the extension behind that statement.

Remark 8 (An Audience Cost on the Operation): Suppose using the operation imposes an audience cost $a_L \in [0, a]$ after either target response, in addition to the operating cost λc_1. After resistance, $\kappa_L(0) = \frac{z+a-a_L}{\lambda}$ and $\kappa_W(0) = \frac{p-z+a_L}{1-\lambda}$, so the operation is active, and lowers the probability of war, if and only if $z + a - a_L > \lambda(p + a)$. After compliance, the cutoffs on the active branch are $\kappa_L(x) = \frac{z-a_L}{\lambda}$ and $\kappa_W(x) = \frac{p-x-z+a_L}{1-\lambda}$, so the operation is active if and only if $a_L < z - \lambda(p - x)$; on that branch $\alpha = 1 - F\left(\frac{z-a_L}{\lambda}\right)$, and once the inequality fails, the operation is dominated after compliance and $\alpha = 1 - F(p - x)$ instead. On the active closed-form branch, which requires $a_L \leq z + a - \lambda\bar{c}$ to maintain $L(0) = 1$, the binding acceptance condition is (19), and x^* and Ω^* are strictly increasing in a_L when $z < h$. Setting $a_L = 0$ recovers the baseline.

$$x^2 + x((1 - \lambda)\bar{c} - p + 2z - a_L - h) - (1 - \lambda)\bar{c}z\alpha = 0 \tag{19}$$

The cutoffs follow from the payoffs. Settlement after resistance still pays $-a$, while the operation pays $z - \lambda c_1 - a_L$, which gives the two post-resistance cutoffs; the condition $\kappa_W(0) < \kappa_L(0)$ reduces to $\lambda(p + a) < z + a - a_L$, and comparing $\kappa_W(0)$ with $\kappa_S(0)$ gives the same inequality, so entry and war reduction again coincide. After compliance the transfer is in place and no audience cost attaches to settlement: settlement pays x, the operation pays $x + z - \lambda c_1 - a_L$, and war pays $p - c_1$, which gives the post-compliance cutoffs and both branches of α.

In the active closed-form region, $L(x) = \frac{z-a_L}{\lambda\bar{c}}$ and $L(0) = 1$, while $W(0) - W(x) = \frac{x}{(1-\lambda)\bar{c}}$ because a_L shifts both war cutoffs equally. The substitution used for Proposition 4 gives $\Omega^* = \frac{x^*(h-z)}{(1-\lambda)\bar{c}} + z\alpha$ at the positive root of (19). Implicit differentiation gives

$$\frac{\partial x^*}{\partial a_L} = \frac{x^* + (1 - \lambda)\frac{z}{\lambda}}{2x^* + (1 - \lambda)\bar{c} - p + 2z - a_L - h} > 0$$

so Ω^* rises with a_L on this branch when $z < h$. Once the operation is dominated after compliance, neither (19) nor its formula for α applies; the comparison follows the war–settlement branch and no global ordering follows from the active-branch derivative.

F.4. The Closed-Form Region

Beyond Assumption 1, Proposition 2′s closed form requires

$$\lambda(p - x^*) < z < p - x^*, \quad \lambda\bar{c} - a \leq z < \lambda\bar{c}, \quad z \leq \frac{1 - p - c_2}{2}$$

The first pair makes the operation active after compliance and keeps the post-compliance war cutoff interior. The second pair combines $\kappa_L(0) \geq \bar{c}$ with $\alpha > 0$ and implies

$$\alpha \leq \min\left\{1, \frac{a}{\lambda\bar{c}}\right\}$$

so the probability of settlement after compliance is capped strictly below one whenever $a < \lambda\bar{c}$. The condition $\kappa_L(0) \geq \bar{c}$ also sets $L(0) = 1$: no coercer type settles after resistance, so the post-resistance menu is effectively binary while the post-compliance menu is ternary. Every closed-form result therefore describes a region in which backing down never occurs, which is the asymmetry that makes Θ^* large and the truncation in (14) one-sided.